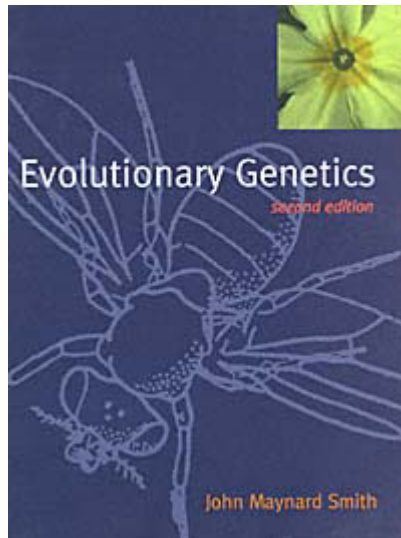




Evolutionary Genetics

Second Edition

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Preface to the Second Edition

The main difference between this and the first edition is the addition of a final chapter on the use of molecular data for the construction of phylogenetic trees. I have done this in response to suggestions from teachers who have used the book as a course text. There are, of course, several excellent computer packages into which one can, more or less mindlessly, plug one's molecular data, and recover a tree with mysterious 'bootstrap values' attached to it. I think it is important, therefore, that biologists should understand the logic underlying these packages, and this I have tried to explain. But I do urge them to remember that molecular data can be used to answer questions about the mechanisms of evolution, as well as about phylogeny.

I have also taken the opportunity to rewrite some sections that students have found confusing. The two chapters that seem to have caused most difficulty are those on the evolution of sex, and on evolutionary game theory. It is ironic that these are the topics on which I have concentrated my own research: perhaps I am too close to them to see the difficulties. In any case, I have rewritten both chapters, and hope that they are now easier to follow.

In general, the discussion of current areas of research in the first edition has stood the test of time rather well. I have expanded some sections, in particular those on the evolution of prokaryotes, and on parasitism and mutualism. Finally, I have corrected a few errors that crept into the first edition, for which I apologize.

J.M.S.

SEPTEMBER 1997

Preface to the First Edition

Ever since Darwin, the theory of evolution has been the main unifying idea in biology. It is natural selection that has made biological systems different from physical or chemical ones. Today, there is an increasing tendency for biology students to specialize either in molecular and cellular biology, or in the biology of whole organisms and populations. Some such specialization is perhaps inevitable, because no one can know everything: it is in any case better than the old division into botanists and zoologists. A course in evolution, however, should unite both streams. Much of molecular biology makes sense only in the light of evolution: the techniques of molecular genetics are essential to a population biologist.

This book is intended as a text for advanced undergraduates: I hope it will also be useful to graduate students. It aims to do two things. First, it provides a basic grounding in those aspects of genetics, both population and molecular, that are needed to understand the mechanisms of evolution. Secondly, it discusses a range of topics, from the evolution of plasmids and of gene families to the evolution of breeding systems and of social behaviour, upon which current research in evolution is mainly concentrated, and attempts to show how the basic principles discussed in the first part of the book can be applied. I am convinced that a proper training in science requires that undergraduates are confronted by the problems of contemporary science. Only then can they see science as an activity, and not as a body of received doctrine. In discussing contemporary problems, I have expressed my own point of view, but I have also given references in which alternative views are expressed.

This is a book about the mechanisms of evolution. It does not describe the techniques, molecular, biometric, or cladistic, whereby phylogenies can be reconstructed. It discusses palaeontology only to the extent needed to ask whether the fossil record demonstrates the existence of mechanisms, such as species selection, other than those deduced from a study of existing organisms.

Further Reading, References, and Definitions

At the end of each chapter, I give a short list of further reading. I have not attempted to give a complete list of references. There is an excellent bibliography of population genetics in Crow and Kimura (1970). I have, however, given references to particular sets of data quoted in the text, and of some classic papers: these are listed at the end of the book.

A number of technical words and phrases are printed in bold type when they first appear, and a short definition is given: the page numbers in the index referring to these definitions are also in bold type.

Some mathematical derivations, and additional factual materials on particular topics, have been set aside from the main text in boxes. You do not need to read the boxes to follow the main text, but some of the problems at the end of the chapters require that you do so.

Problems

The problems at the end of the chapters are an integral part of the book. Solving problems is the only way to learn population genetics. Answers, and an outline of how they were obtained, are given at the end of the book. If you get a different answer, you may be mistaken, or I may be mistaken, or there may be an ambiguity in the question. Obviously, I have tried to avoid the last two possibilities, but I cannot be sure that I have succeeded. I suspect that you will find the problems, or some of them, difficult, but I hope that you will enjoy doing them. Remember that you cannot expect to know the answer to a problem instantly, or merely by looking up the relevant page in the text: it may take time and effort. Those that require more mathematical skill, or extra knowledge, are marked $\star$. Some are open-ended, in the sense that they do not have a unique correct answer: this should be obvious from the question.

Computer Projects

At the end of most chapters, I have suggested a few computer projects. All these (with one exception that is indicated) can be carried out in BASIC on a micro-computer. I have used many of them as final-year assessment exercises for undergraduates studying population genetics at Sussex. For a student with little previous programming experience, a project should take up to six weeks to complete and write up, assuming the student is also attending lectures and practicals. I have sometimes stated that a problem is tricky to program: beginners should steer clear of them. Students without previous programming experience will need a fair bit of help to get started, and most students need some help in formulating the basic model.

Some of the projects are aimed at solving problems that can be solved analytically. This is not as silly as it sounds. Most theoreticians nowadays check their results by simulation, or use simulation to suggest results that might be provable analytically. Also, if you write a program to solve a problem that cannot be solved analytically, it is essential to check the program by running some special cases (e.g. a case with no selection) whose results are known analytically: otherwise there is no way of being sure that the program is doing what it is intended to do.

Background Knowledge

I have assumed some knowledge of genetics, mathematics, and statistics, as follows.

Genetics

Mendelian genetics, the chromosome theory and the nature of meiosis, sex-linked inheritance, the meaning of recombination in classical genetics. The structure and role of DNA, RNA, and protein as described in an elementary biology text. I have not assumed a knowledge of parasexual processes in prokaryotes (transformation, transduction, transposition), or of the nature and behaviour of reiterated DNA in eukaryotes: these matters are described in the text.

Mathematics

Elementary algebra, the manipulation of symbols, and the solution of simple equations. The use of x - y coordinates. The meaning of dx/dt as a rate of change. I have not assumed a knowledge of integration, how to solve differential or difference equations, partial differentiation, or of matrix algebra: but a knowledge of these topics would be of great value if you plan to pursue evolutionary genetics further. But to paraphrase Mr Truman, if you can't stand algebra, keep out of evolutionary biology.

Probability and Statistics.

The first requirement for a population geneticist is an ability to calculate probabilities. Plenty of practice in doing this is provided by the problems at the ends of chapters. But I do assume you know how to use the concept of probability. The following ideas are made use of in the text (usually with a brief explanation): the binomial theorem of probabilities, the Poisson distribution, the normal or Gaussian distribution, the meaning of statistical significance, the χ^2 test, means, variances, covariances, and regression. Clearly, therefore, it would be well to attend a course in probability and statistics before reading this book.

One final word. Forty years as a biologist, and five years before that as an engineer, have convinced me that the main difficulty one faces in a subject like population genetics (or mechanics) is not the mathematics itself, or the biology itself: it is how to fit them together. The only way one can learn to make useful models of the world, whether one is designing an aeroplane or studying the evolution of altruism, is by doing it: in practice, that means by solving problems. The problems and computer projects are intended to help you to acquire the necessary skills.

J.M.S.
UNIVERSITY OF SUSSEX
6 APRIL 1988

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To Carol and S^関n

Chapter 1— Evolution by Natural Selection

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Darwin's Theory

In *The Origin of Species*, Darwin argued that all existing organisms are the modified descendants of one or a few simple ancestors that arose on Earth in the distant past. We now know, over 3000 million years ago. He also argued that the main force driving this evolutionary change was natural selection. The argument is as follows. Living organisms multiply, and would increase indefinitely were not their numbers limited by death. Organisms also vary, and at least some of the variation affects their likelihood of surviving and reproducing. Finally, organisms have the property of 'heredity': that is, like begets like. The essential feature of heredity is illustrated in Fig. 1.1. Notice that heredity can be defined only for entities that both multiply and vary. We do not think of a rock, which is the same today as it was yesterday, as having heredity, because it does not multiply. But multiplication and variation are not sufficient. Fire multiplies, provided that fuel is supplied, and it varies, but it does not have heredity, because the nature of a fire depends on its present 'environment'—fuel, wind, etc.—and not on whether it was lit by a match or a cigarette lighter.

Darwin, then, argued that organisms do in fact multiply and vary, and that they have heredity, and that, in consequence, populations of organisms will evolve. Those organisms with characteristics most favourable for survival and reproduction will not only have more offspring, but will pass their characteristics on to those offspring. The result will be a change in the characteristics present in the population. The evolutionary change does not require that any individual should change, although it does require that new variants arise in the process of reproduction, because otherwise the essential variability of the population would disappear.

The theory of natural selection not only predicts evolutionary change: it also says something about the kind of change. In particular, it predicts that organisms will acquire characteristics that make them better able to survive and reproduce in the environment in which they live. That is, it predicts the adaptation of organisms.

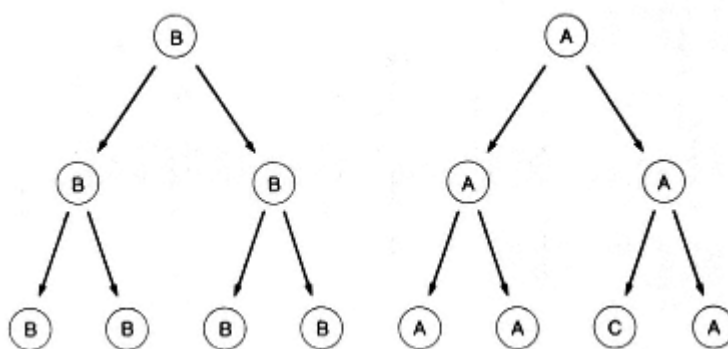


Figure 1.1

Heredity and variation. The meaning of heredity is that, when multiplication occurs, like gives rise to like: A gives rise to A, and B to B. Variation requires that this rule is occasionally broken, as when A gives rise to C.

to their environments. Of course, Darwin was well aware that organisms are adapted before he thought of his theory: adaptation is the most obvious and all-pervasive feature of living things, and one that any theory of evolution must explain. One of the main strengths of Darwin's theory is that it does explain adaptation: as we shall see, its only serious rival, the Lamarckian theory, cannot do so.

There are, however, obvious inadequacies in the theory illustrated in Fig. 1.1. In particular:

1. The figure defines heredity, but says nothing about its mechanism. In fact, organisms are not replicated in the process of reproduction. They die, and only their gametes are passed on. Modern genetic theory asserts that the only thing that is exactly replicated is the information in the DNA (or, in some viruses, the RNA): other structures must develop anew in every generation. (Some possible exceptions are discussed below.)
2. The figure implies that each individual has only one parent. In higher organisms, biparental sexual reproduction is typical, although not universal. Even in prokaryotes, DNA from different ancestors may come together in a single descendant.

In brief, Fig. 1.1 ignores the phenotype-genotype distinction, and it ignores sex. A large part of this book is concerned with these two complicating factors. First, however, I discuss some experiments in which sex is absent, and in which the distinction between genotype and phenotype, although not wholly absent, is minimal. These experiments concern the evolution of RNA molecules *in vitro*.

Evolution *in vitro*

There is an RNA virus, Q β , that infects the bacterium *Escherichia coli*. The virus genome codes for an enzyme β replicase that replicates RNA. The enzyme

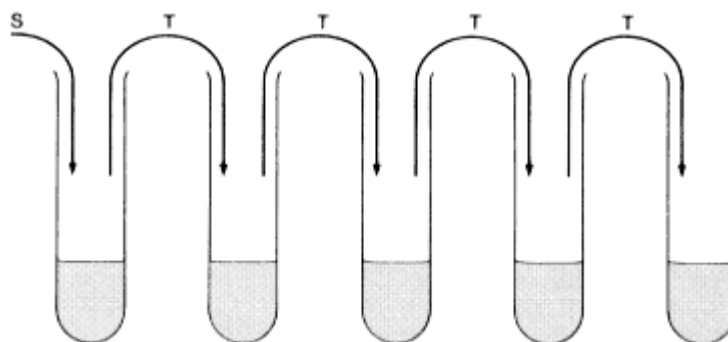


Figure 1.2

The evolution of RNA molecules *in vitro*. Initially, each test tube contains a solution of the four nucleotides ATP, GTP, UTP, and CTP from which RNA is made, and an enzyme that will replicate RNA. RNA molecules are added as a seed (S) to the first tube. After 30 min, a drop of solution is taken from the first tube, and added to the second (T); after a further 30 min, a drop is taken from the second tube, and added to the third, and so on.

works well *in vitro*, and will replicate almost any RNA molecule in a test-tube, if it is provided with the four necessary monomers from which RNA is made—ATP, GTP, UTP, and CTP. Hence one can follow the evolution of a population of RNA molecules *in vitro*. The experimental system is shown in Fig. 1.2. A primary RNA template is added to a test-tube containing Φ replicase and the four monomers. After about 30 min, a small fraction of the contents of the tube is withdrawn and added to a second tube: this process can be repeated for 100 or more transfers.

If replication was exact, the RNA molecules present after 100 transfers would be identical to the original template. But replication is not exact. The probability that a 'wrong' base—that is, one not complementary to that in the strand being copied—will be incorporated is about 1 in 10 000, per base, per replication. Other errors also occur, when part of a strand is not copied at all (deletion), or is copied twice (duplication). There is therefore variation upon which selection can act. But why should one RNA strand be better or worse than another? There are two reasons. One rather boring reason is that, within limits, short strands are replicated faster than long ones. A more interesting reason is that RNA molecules have a three-dimensional structure, because a molecule bends back on itself, forming hairpin-like structures held together by pairing between complementary bases. This is illustrated in Fig. 1.3, which shows the secondary structure of a molecule 218 bases long which, because of its secondary structure, is replicated particularly rapidly by Φ replicase.

Experiments of the kind shown in Fig. 1.2, then, ought to lead to Darwinian evolution, and they do. After a number of transfers, the initial template molecules are replaced by a population of molecules, similar or identical to one another, and replicating much more rapidly. Of particular interest are experiments in which no initial template molecules are added. One might then suppose that, with nothing

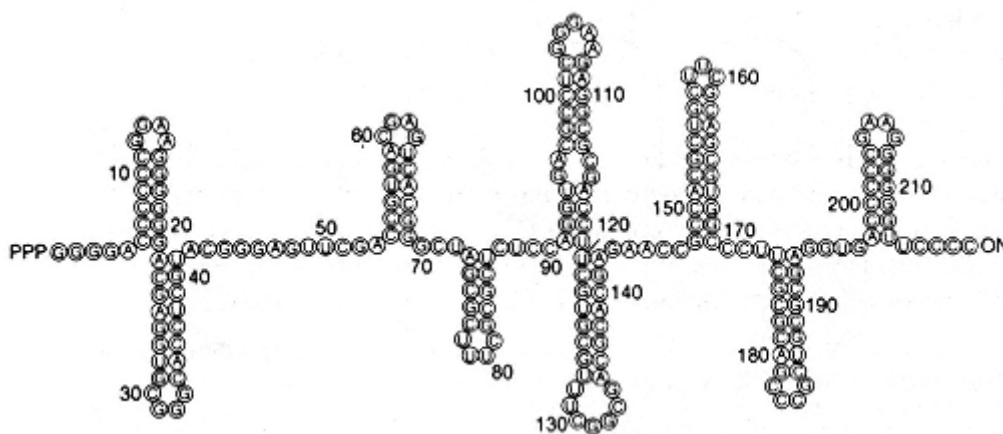


Figure 1.3
An RNA molecule that evolved *in vitro*. (From Orgel 1979.)

for the enzyme to copy, nothing would happen. However, after a substantial time delay, very short RNA templates, consisting of only a few nucleotides, do appear, and their length increases in subsequent transfers. (There is some controversy about whether the initial oligomers really appear *de novo*, by linking monomers, or whether they are present as impurities, but this is unimportant in the present context.) Evolutionary change finally comes to a halt. The nature of the final population depends on experimental conditions—for example, ionic composition of the medium, and presence of inhibitory drugs. For any particular set of conditions, however, the length and sequence of the final population is repeatable. The molecule in Fig. 1.3 is one such end-point. It also closely resembles a molecule, known as a minivariant, that is found naturally in *E. coli* infected by the Q β virus. How does this minivariant come to exist in nature? It could not multiply by itself in *E. coli*, if only because it does not code for a replicase. However, if a cell is infected by a functional Q β virus, the minivariant can exist as a kind of super-parasite, relying on the replicase coded for by the virus, which itself relies on many enzymes coded for by the host bacterium. The *in vitro* experiment repeats, in a test-tube, the evolutionary process that gives rise to the minivariant in nature.

The first moral to be drawn from these experiments is that natural selection can produce highly improbable results. There are 4^{18} , or 10^{128} , different RNA molecules 218 bases long. The one illustrated in Fig. 1.3 is unique in being the one replicated most rapidly by Q β replicase in the conditions of the experiment. How have we been able to produce this one unique sequence so quickly? Thus, there are approximately 10^6 RNA molecules in a test-tube just before transfer. After 100 transfers, we have tried out at the most 10^8 molecules. We seem to have been very lucky to have hit the optimal sequence so soon. If we could look at 10^6 molecules every half hour, each one different from every other, it would take 10^{107} years to have a reasonable chance of finding the uniquely best one.

It is a fallacy to imagine that natural selection works by trying out, at random, all possible phenotypes until, by chance, it hits on the best one. Instead, natural selection is a process analogous to hill-climbing, in which the best phenotype is reached by a series of steps, each step leading to a type that is fitter than the previous one (the precise meaning of 'fit' is discussed in Chapter 3). Applied to the *in vitro* experiment just described, this concept of hill-climbing implies the following. The process started with a short random sequence A, and ended with a unique sequence Z that is replicated particularly rapidly. For this to happen, there must be a series of intermediates, A-B-C-...-M-N-...-Z, such that:

1. Each step or example, M-N can arise by a single mutation that is, a base substitution, deletion, or duplication.
2. Each step increases replication rate. There could be some debate about whether a few of the steps could be 'neutral', in the sense of neither increasing nor decreasing replication rate, but the calculations in the last paragraph show that if most steps were neutral we would never arrive at Z.
3. The total number of steps is not very great. Note that, by base substitution, one can travel from any RNA molecule n bases long to any other of the same length in a maximum of n steps, although there is no guarantee that all the steps would be improvements.

If these conditions hold, the population will evolve from A to Z reasonably quickly. The fact that the *in vitro* experiments do repeatedly lead to the same end-point can be taken as evidence that, in this case, the three necessary conditions do hold. However, it is worth noticing that the end-point or example, the molecule of Fig. 1.3 may not be, as implied above, the uniquely best sequence. Thus it may be that, starting from A, there is an uphill path to Z, but that there is some other molecule, say OPT, which is replicated even more rapidly than Z, but which cannot be reached by hill-climbing from A, because to reach OPT would require the simultaneous incorporation of several mutations, each by itself deleterious.

These *in vitro* experiments, then, do illustrate the power of natural selection to generate the improbable. However, they have limitations as models of evolution. First, it is in a way disappointing that evolutionary change comes so quickly to a halt. In the real world, evolution seems to continue indefinitely. What is needed if this is to be so? This question is harder than it looks: it will be discussed briefly in the last chapter. A more immediate limitation lies in the absence of a clear distinction between phenotype and genotype, and of a process of development. In a sense, the genotype of an RNA molecule is its base sequence, and its phenotype is its three-dimensional structure. The analogue of development is then the process of folding. This is correct, but the situation is too simple to provide an adequate background for discussing the main alternative to Darwinism, which is the theory commonly referred to as Lamarckism, discussed in the next section.

The point of describing these *in vitro* experiments is to illustrate three fundamental ideas:

1. A population of entities (in this case, molecules) that have the properties of multiplication, variation and heredity will evolve so that they are better adapted to survive and reproduce.
2. This process of natural selection can give rise to structures whose probability of arising by chance in a single step is vanishingly small.
3. The process is analogous to hill-climbing. It doesn't work if there is no hill to climb: that is, if there is no series of intermediate steps leading to the summit.

Lamarck, Weismann, and the Central Dogma

The theory that today goes under the name of Lamarckism is a much modified version of the views of the French biologist Lamarck (1744-1829). We cannot simply dismiss this theory as false, for two reasons. First, it is not so obviously false as is sometimes made out. Secondly, it is the only alternative to Darwinism as an explanation of the adaptive nature of evolution. The idea is as follows. During its life, an organism may adapt to its environment. The classic, and convenient, example is that a blacksmith develops arm muscles appropriate to his trade. Other examples are that humans living at high altitudes produce more red blood cells, that humans acquire immunity to diseases to which they are exposed, and that they learn to drive on the correct side of the road. All these changes make them better able to survive, and all are responses to a particular environment during an individual lifetime. If this kind of adaptation is to be relevant to evolution, the changes that occur in an individual must have some effect on the nature of its offspring. If they do, this will contribute to the evolution of new and improved adaptations.

Darwin accepted this possibility, under the term 'the effects of use and disuse', although he thought that natural selection was a more important cause of evolution. When he said that he rejected Lamarck's views, it was not this idea he was rejecting, but Lamarck's belief that organisms have an inherent drive to evolve into higher and more complex forms. Darwin saw, correctly, that to explain the evolution of complexity in this way is like explaining the fact that the universe is expanding by saying that it has an inherent tendency to get bigger. The Lamarckian theory of the inheritance of acquired characters was explicitly rejected by August Weismann (1834-1914). He claimed (Fig. 1.▲) that, starting from the fertilized egg, there are two independent processes of cell division, one leading to the body or 'soma', and the other the 'germ line' leading to the gametes that form the starting point of the next generation. Of these two cell lines, the soma will die, but the germ line is potentially immortal.

Weismann's central claim was that the germ line is independent of changes in

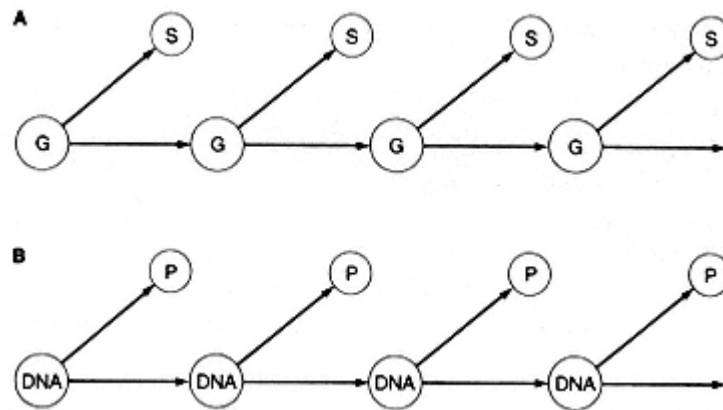


Figure 1.4
Weismann and the central dogma.

the soma. If this is true, then acquired characters cannot be inherited. But it is not clear why he thought it was true. He did point out that in most animals—for example, vertebrates and insects—the primordial germ cells that will give rise to the gametes are set aside early in development. This is true enough: if the primordial germ cells are absent, or are destroyed, they cannot be replaced, and the animal is sterile. However, this does not prove Weismann's point, for two reasons. First, in higher plants there is no independent germ line: any cell in a growing shoot can give rise to gametes. Yet the non-inheritance of acquired characters is held to be as true of plants as of animals. Secondly, the energy and material needed for the production of gametes are provided by the rest of the body, so there are opportunities for the soma to influence the germ line. In fact, Weismann's insight was to realize that what is relevant is the passage, not of material or energy, but of information. In effect, he could not see how the large muscles of a blacksmith could so influence the sperm he produced that his sons would develop large muscles. That Weismann saw that the problem is one of information transfer is shown by his remark 'If one came across a case of the inheritance of an acquired character, it would be as if a man sent a telegram to China, and it arrived translated into Chinese.'

Today, we would express Weismann's argument in molecular terms. Figure 1.4B shows the 'central dogma' of molecular biology, which asserts that information can pass from DNA to DNA, and from DNA to protein, but not from protein to DNA. By 'information' is meant the base sequence of DNA, which is transmitted to new DNA molecules in the process of replication, and which specifies the amino-acid sequence of proteins in the process of translation.

It is important to be clear about what is being asserted by the central dogma. It is not true that DNA can replicate without proteins: enzymes are needed. Further, changes in enzymes can alter the way in which a particular DNA sequence is translated. What does seem to be true, however, is that, if a protein with a new

amino-acid sequence is present in a cell, it cannot cause the production of a DNA molecule with the corresponding base sequence. Notice that this is not a logical necessity. Machines that translate information can sometimes work both ways: a tape recorder can translate sounds into magnetic patterns on a tape, and vice versa. But some machines translate only in one direction: you cannot cut a record by singing into the speaker of a record-player. The central dogma claims that the relation between nucleic acids and proteins resembles a record-player, and not a tape recorder.

The fact that information passes from DNA to protein through an RNA intermediate complicates the argument, but does not alter the essentials. There are RNA viruses that code for an enzyme reverse transcriptase that can copy RNA base sequences into DNA. This means that the flow of information is as in Fig. 1.5.

If the central dogma is true, and if it is also true that nucleic acids are the only means whereby information is transmitted between generations, this has crucial implications for evolution. It would imply that all evolutionary novelty requires changes in nucleic acids, and that these changes are essentially accidental and non-adaptive in nature. Changes elsewhere in the egg cytoplasm, in materials transmitted through the placenta, in the mother's milk might alter the development of the child, but, unless the changes were in nucleic acids, they would have no long-term evolutionary effects. The rest of this book is based on the assumption that this neo-Darwinist picture is correct. But first, I review some contexts in which the assumptions are dubious, or actually false.

1. *Cell differentiation.* The cells of higher organisms are differentiated for example, fibroblasts, epithelial cells, leucocytes, and so on. The differences between these cells are hereditary, in the sense defined in Fig. 1.1; that is, they are stable through many cell divisions. However, with a few exceptions (e.g. in the immune system), the differences are not caused by differences in DNA

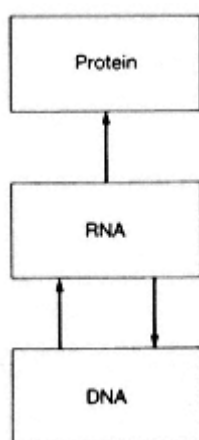


Figure 1.5
The flow of information in the genetic system.

base sequence, but by different states of activation of genes. Typically, these different states are abolished (or were absent in the germ line) when gametes are produced. However, we cannot rule out the possibility that some changes in gene activation might be transmitted in sexual reproduction. The members of a clone of *Daphnia* can have different morphologies: for example, they develop spines in the presence of predators. The change in morphology is adaptive; it occurs in response to an environmental stimulus; and once it has occurred, it is transmitted through the egg. Almost certainly, it is caused by changes in gene activation and not by changes in the base sequence.

2. *Changes in gene amplification.* Perhaps the clearest example of Lamarckian inheritance occurs in flax (*Linum*). If flax plants are treated with high levels of fertilizer, their morphology changes (Cullis 1983). These changes persist for a number of sexual generations (although not indefinitely) in the absence of the fertilizer treatment. It turns out that, in the cells of the modified plants, some DNA sequences (including ribosomal genes) are present in a higher number of copies. Thus the changes involve gene amplification, but probably not the appearance of new sequences.

3. *Cortical inheritance in ciliates.* The surfaces of ciliated protozoa contain complex patterns of cilia. If the pattern in an individual is changed, either accidentally or by surgical interference, the new pattern may be transmitted through many binary fissions. This transmission occurs independently of any change in nuclear DNA. It seems that there is a second hereditary mechanism, not dependent on nucleic acids, and subject to Lamarckian effects: a possible mechanism is described by Sonneborn (1970). It is not known whether any comparable mechanism exists outside the ciliates.

4. *Cultural inheritance.* If an animal learns where the water-holes are, or what plants are safe to eat, this information may be transmitted to its offspring, and to more distant descendants. In our own species, cultural inheritance is the basis of historical change.

To summarize, the strict assumptions of neo-Darwinism are contradicted by transmissible states of differentiation, by transmissible gene amplification, and by the existence of alternative hereditary mechanisms (cortical inheritance, cultural inheritance) not dependent on nucleic acids. How does this affect evolution theory?

Much the most important modification arises from cultural inheritance, because the traits that are acquired during a lifetime and then transmitted are often adaptive in nature: an animal that knows which berries are edible is more likely to survive. Given sufficient capacity for learning and cultural communication, a population can adapt to its environment by non-genetic means. The mechanisms of history and of evolution are so different that it is best to distinguish clearly between them. However, they may interact.

Other alternative hereditary mechanisms, and in particular cortical inheritance, are of less importance, because they are not adaptive: that is, the change occurring in an individual's life does not, in general, improve survival.

The significance of the experiments with flax is harder to evaluate. It is not clear that the morphological change adapts the plant to increased fertilizer, but it may well do so. If so, we are looking at an adaptation of the genetic system itself, enabling a parent plant to produce seedlings adapted to a changed environment. Until we know more of the molecular mechanisms involved, it is hard to decide how common processes of this kind may prove to be. However, if the morphological change is indeed adaptive, the genetic system responsible for the gene amplification and its transmission must itself have evolved by natural selection.

There remains the question why Lamarckian inheritance is not more common than it is: the examples given above are the best there are, and they are atypical. The short answer is that Lamarckian inheritance is rare because the central dogma is true, and because nucleic acids are overwhelmingly the most important carriers of genetic information. That, however, is to give an explanation in terms of genetic mechanisms. We would like also to know why the genetic mechanism is like that. If a tape recorder can be designed to transmit information in both directions, surely a genetic mechanism with a two-way flow of information from phenotype to genotype as well as from genotype to phenotype could have evolved. The answer is that most phenotypic changes (except learnt ones) are not adaptive: they are the result of injury, disease, and old age. A hereditary mechanism that enabled a parent to transmit such changes to its offspring would not be favoured by natural selection.

Further Reading

Darwin, C. (1859). *On the origin of species*. Murray, London.

Problems

The following problems are based on Dawkins' (1986) computer model of evolution by natural selection. They provide good practice in calculating probabilities. They also illustrate the power of selection to generate highly improbable results. (Two hints about calculating probabilities. First, if you can't calculate the probability that something will happen, calculate the probability that it won't. Secondly, a useful approximation: $(1 - x)^n \cong e^{-nx}$, if x is small and n is large.)

Dawkins models the evolution of the message 'METHINKS IT IS A WEASEL'. For simplicity, ignore spaces, and let the 'correct' 19-letter message be 'METHINKSITIS. AWEASEL'. Start with a single 19-letter message, in which each letter is randomly chosen from the 26 letters.

1. How many such messages are there?
2. What is the probability that (a) at least one of the 19 letters is correct, (b) exactly one letter is correct?

Ten copies are made of the original message. In copying each letter, there is a chance of 99/100 of incorporating an unchanged letter, and of 1/100 of incorporating a changed letter 桎 `mutation' 桎 which may, with equal probability, be any one of the other 25 letters.

1. Suppose that, in the original sequence, none of the letters matched the correct message. What is the probability that, in at least one of the 10 copies, at least one letter does match the message?

The best of the 10 copies (that is, the copy that matches the required message at the largest number of sites) is chosen as the `parent' of the next generation: if two or more copies match at the same number of sites, one is chosen at random. This parent is used to generate 10 more copies, in the same way.

4. If the original sequence did not match the correct message at any sites, approximately how many generations will pass before a message matching at least one site is obtained?

5. Sooner or later, a message correct at 18 out of 19 sites will be obtained. What is the probability that, among the 10 copies of such a message, one will be correct at all 19 sites?

6. What, in your opinion, is the least realistic feature of this model, regarded as a model of evolution by natural selection?

Computer Projects

Simulate Dawkins' `METHINKSITISAEAL' model of evolution, as described in the problem above. How long does it take to evolve from 0 to 19 correct letters? Modify it to allow for recombination. For example, keep two sub-populations of n messages. Once every r generations, take the best message from each population, and generate two new sub-populations by recombination between them. Does such a population evolve faster than a single population of $2n$ individuals, from which two are selected every generation?

Chapter 2— Models of Populations

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The *in vitro* experiments described in the last chapter demonstrated that evolution occurs in a population of entities that have heredity. In this chapter, I develop some simple models of population growth and evolutionary change, aimed at answering the following questions:

1. In what circumstances will one type replace another selectively?
2. Can selection lead to the evolution of two different but coexisting types?
3. How accurate must the hereditary process be?

Models of Population Growth.

Imagine a population growing asexually, by binary fission. For example, a population of bacteria. Suppose that, at time t , the population contains x individuals, and that we can watch one of these for a short time δt . Let the probability that it will divide during that time be $r\delta t$: for the present, assume that r is constant. Then the increase in the number of individuals is

$$\delta x = r\delta t x. \quad (2.1)$$

As $\delta t \rightarrow 0$, this equation can be replaced by the differential equation,

$$dx/dt = rx, \quad (2.2)$$

of which the solution is

$$x = x_0 e^{rt}, \quad (2.3)$$

where x_0 is the number of individuals at time $t = 0$.

Taking logarithms of both sides, this becomes

$$\ln x = \ln x_0 + rt. \quad (2.4)$$

That is, if we plot the natural logarithm of the number against time, we get a straight line with slope r which is called the intrinsic rate of increase. Because of this

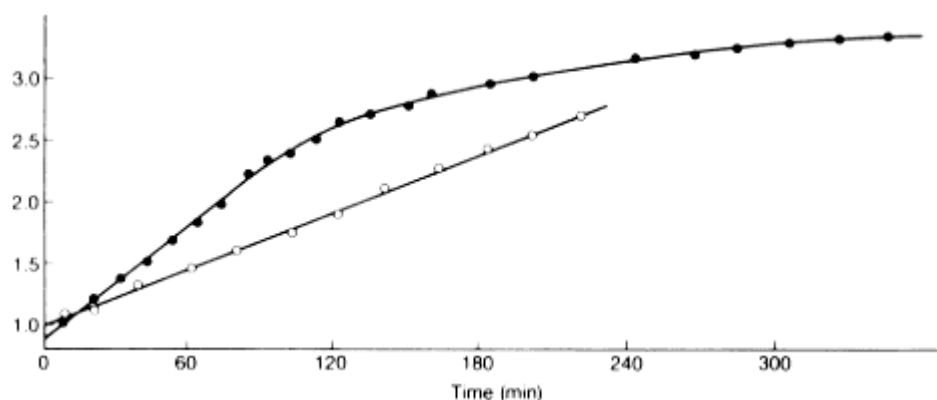


Figure 2.1

Growth of two cultures of *E. coli*. Closed circles, in nutrient broth; open circles, in synthetic medium. Turbidity is proportional to number of cells per unit volume. (From Stent 1963.)

linearity, growth obeying Equation 2.4 is called logarithmic growth. Figure 2.1 shows that bacterial populations do, for a time, obey this law.

Will r in fact be constant? There are several reasons why it might not be. For example:

1. *Synchrony*. Suppose we start with a population of bacteria which had all just completed division. Then, if the inter-division time was, say, 30 min, there would be no divisions for that period, and then all the cells would divide almost simultaneously. Thus r would not be constant, and the population would grow in a stepwise manner. If the division time was exactly 30 min for all cells, the synchrony would last for ever. In practice, however, there would be some variation either side of 30 min, and in time the synchrony would be lost. There is a theorem, due to Lotka, which says that if there is a population whose members have age-specific birth and death rates—that is, which have a fixed probability both of reproducing and of dying as a function of age, then in time that population will reach a stable age distribution: after that, the proportion of the population that is of any age remains constant. In our model, the death rate is zero, and the birth rate is fixed, with a narrow peak at 30 min. Lotka's theorem says that in time our population will reach a stable age distribution. At any instant, some cells will just have divided, and others will be about to divide, but, if we choose a cell at random, its chance of dividing will be constant. This justifies Equation 2.1 as a representation of a population in a stable age distribution.

2. *Inherited differences between cells*. Suppose that there are cells with different division times, and that these differences are transmitted to daughter cells. Then the proportions of different kinds will change with time, and r will not be constant. Thus Equation 2.4 assumes no inherited differences.

Even if the population is asynchronous and genetically uniform, it cannot increase logarithmically for ever. Sooner or later, a shortage of resources must bring the increase to a halt. It was this insight that both Darwin and Wallace acquired by reading the economist Malthus, and which led them to the idea of natural selection. The effect of resource limitation is allowed for in the logistic equation:

$$dx/dt = rx(1 - x/K). \quad (2.5)$$

Although this equation is the basis of much of theoretical ecology, it has weaknesses, which are discussed in Box 2.1. It does, however, have two essential features, which make it adequate for our present needs:

1. When x is small, it reduces to Equation 2.2: this corresponds to the fact that populations, when small, often increase exponentially.
2. Population growth slows down, and reaches an equilibrium level, K (see Fig. 2.2). K is called the carrying capacity. If, initially, $x < K$, the population rises towards K , and if $x > K$ the population falls towards K . Note that the approach to K occurs without oscillations. This does not mean that fluctuations in population size cannot occur in the real world, but only that Equation 2.5 was chosen because it is a convenient description of populations that do not oscillate.

Selection in an Asexual Population

We are now in a position to ask our first question: in what circumstances will selection lead to the replacement of one kind of organism by another? Suppose that the numbers of the two kinds are x and y , respectively. Then we might describe their growth by the equations:

$$\begin{aligned} dx/dt &= r_1x(1 - x/K_1), \\ dy/dt &= r_2y(1 - y/K_2). \end{aligned} \quad (2.6)$$

It is then easy to see what will happen: x will increase to K_1 , and y will increase to K_2 . The two populations will coexist indefinitely. There will be no selective replacement of one by the other.

The reason for this rather disappointing result is that, in Equation 2.6, we have assumed that the two populations are limited by different resources. This is implicit in the fact that the growth of each is unaffected by the other. Let us, therefore, make the opposite assumption—that the two kinds have the same resource requirements. That is, we assume that the growth of x is slowed down as much by the value of y as it is by the value of x , and vice versa. Then

$$\begin{aligned} dx/dt &= r_1x[1 - (x + y)/K_1], \\ dy/dt &= r_2y[1 - (x + y)/K_2]. \end{aligned} \quad (2.7)$$

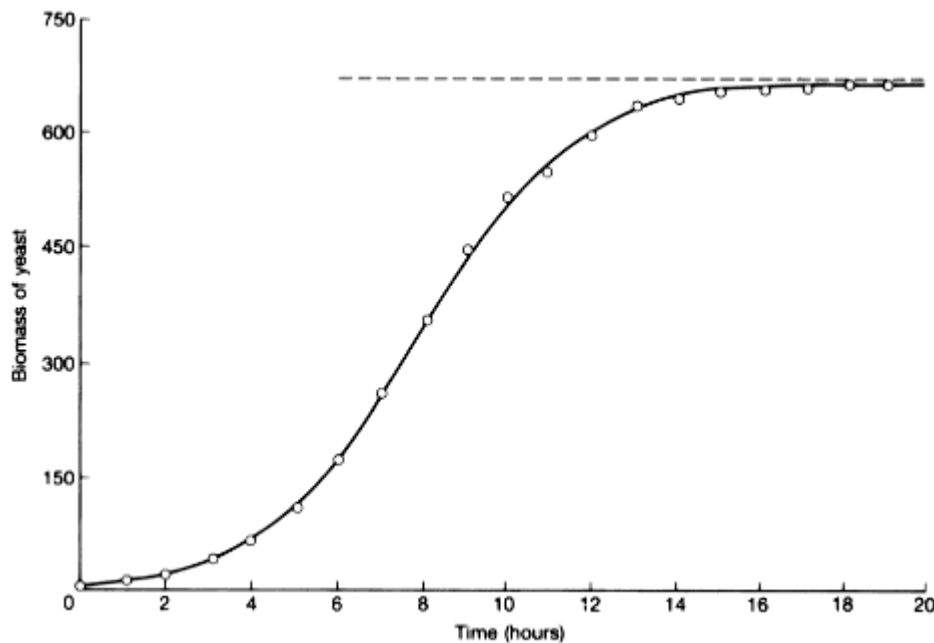


Figure 2.2

A comparison of the growth of yeast in a culture with logistic growth (from Allee *et al.* 1949).

It is now easy to see that, if K_1 and K_2 are different, one kind will eliminate the other. Thus suppose that $K_1 > K_2$. Then x will increase until $x + y = K_1$. At this point, $x + y > K_2$, and hence dy/dt is negative. Thus y will decrease: in fact, y decreases to zero, so that x selectively eliminates y .

The essential point, then, is that natural selection will cause the replacement of one type by another if, and only if, the two are competing for resources, or, more generally, are limited by the same factors. In ecological terms, they must be controlled by the same negative density-dependent factors. In *in vitro* experiments, this is certainly the case: all RNA molecules are competing for the same replicase enzymes, and the same nucleotides.

There is one feature of the conclusion from Equation 2.7 that is misleading. Since x wins if $K_1 > K_2$, and y wins if $K_2 > K_1$, it might seem that only a difference in carrying capacity K , and not in intrinsic rate of increase, r , could lead to selective replacement: in ecological language, it suggests that only traits that affect resistance to density-dependent factors are subject to natural selection. This is an unfortunate feature of the logistic equation: it is shown in Box 2.1 that selective replacement occurs between forms that differ only in intrinsic rate of increase.

In comparing Equations 2.6 and 2.7, we compared a model in which the two types had no limiting factor in common, and one in which the limiting factors were

identical. Box 2.2 deals with the intermediate case, in which the limiting factors are similar but not identical. Two main conclusions emerge:

1. Coexistence is favoured if each kind has a greater inhibiting effect on its own growth than it does on the other.
2. When there is coexistence, each kind has a higher rate of increase than its competitor when it is rare, and its competitor is common.

Usually, the model in Box 2.2 is thought of as applying to competition between species, and as giving the conditions that must be satisfied if two species are to coexist. However, it applies equally well to competition between different geno-

Box 2.1—

A Non-logistic Model of Population Growth and Competition

There is an important difference in kind between Equations 2.2 and 2.5. This can be expressed by saying that Equation 2.2 can be microscopically justified, whereas Equation 2.5 is descriptive and phenomenological. Thus Equation 2.2 derives from a description of what individual cells are doing—for example, that they are asynchronous, genetically identical, and not resource-limited. Equation 2.5 does not. For example, it is possible for dx/dt to be negative (when $x > K$), and therefore there must be deaths as well as births, but it was nowhere stated how births and deaths contribute. The justification for Equation 2.5 is that it accurately describes some cases of change in population number, and that it is mathematically simple. In fact (as indicated in Problem 1) it is possible to derive Equation 2.5 from a consideration of what individuals are doing, but it was presented here as purely phenomenological.

It may help to note that this contrast between phenomenological and microscopically justified theories is common in science. A classic example is the contrast between statistical mechanics, in which the behaviour of a gas is deduced from the behaviour of individual molecules, and classical thermodynamics, which describes systems in terms of global properties—temperature, heat, entropy, etc.—without microscopic derivation. Most theories in biology have both microscopic and phenomenological features.

Let us replace Equation 2.5 by an equation based on the behaviour of individuals. The change in x is equal to the number of births, minus the number of deaths. We can write

$$\delta x = (\text{birth rate} - \text{death rate})x\delta t.$$

If we assume that the death rate is constant, but the birth rate is density-dependent, this becomes

$$\delta x = [r(x) - d]x\delta t.$$

We need a form for $r(x)$. At this point, the model becomes descriptive. We need a form in which $r(x) = r$ when x is small, and tends to zero as x tends to infinity. Algebraically, the simplest form is $r(x) = r/(1 + x)$. Then

$$\frac{dx}{dt} = \frac{rx}{1+x} - dx. \quad (2.8)$$

This equation can be integrated, to give a pattern of growth that would be indistinguishable from logistic growth, with intrinsic rate of increase r , and carrying capacity $K = (r - d)/d$.

Suppose now that two types, x and y , have identical limiting factors. Then

$$\begin{aligned} \frac{dx}{dt} &= \frac{r_1 x}{1+x+y} - d_1 x, \\ \frac{dy}{dt} &= \frac{r_2 y}{1+x+y} - d_2 y, \end{aligned} \quad (2.9)$$

Then $dx/dt = 0$ when $1 + x + y = r_1/d_1$, and $dy/dt = 0$ when $1 + x + y = r_2/d_2$. In competition, that type will win that has the larger value of r/d .

The relevance of this is as follows. It might seem from the analysis of Equation 2.6 in the main text that only differences in carrying capacity, and not of intrinsic rate of increase, could lead to selective replacement. This, however, is a peculiarity of the logistic equation. For the more realistic model of Equation 2.9 differences in either birth or death rate, or both together, can influence the outcome.

types in an asexual population. The conditions for the coexistence of different types in a sexual population are more complex: they are discussed in Chapter 4. However, it is worth noting at this stage that condition 2 above amounts to saying that the fitness of a type depends on its relative frequency in the population, increasing as it becomes less frequent. Thus the stable equilibrium illustrated in Fig. 2.3 is an example of 'frequency-dependent selection', as discussed on page 69.

The Accuracy of Replication

If the replication process were exact, no new variants would arise, and evolution would slow down and stop. The *in vitro* experiments work only because enzyme replication of RNA is not exact. However, evolution would also be impossible if the replication process were too inaccurate. Thus although an occasional error in replication, **omutation**, may lead to an improvement in adaptation, most will lead to deterioration. Hence, too high an error rate will lead to loss of adaptation. I now

Box 2.2—**Competition Between Two Types**

By analogy with Equations 2.6 and 2.7, we can write

$$\begin{aligned}\dot{x} &= r_1x(1 - bx - cy), \\ \dot{y} &= r_2y(1 - fx - gy),\end{aligned}\tag{2.10}$$

to describe the population sizes x and y of two types which inhibit one another's growth, but to a degree different from that with which they inhibit their own. In these equations, I have used the notation $\dot{x}$ and $\dot{y}$, instead of dx/dt and dy/dt , for the rates of change of x and y . The constants r_1 and r_2 are the intrinsic rates of increase of x and y ; b and g measure the inhibiting effects of each type on its own increase, and c and f the inhibiting effects of each type on the growth of the other. In the notation of the main text, $K_1 = 1/b$, and $K_2 = 1/g$.

We can represent the state of the two populations at any moment by the values of x and y . Therefore we can also represent the state as a point in a two-dimensional **state space**. We can then describe changes in the two populations in time by attaching an arrow to each point in state space, indicating the direction in which the system will move. Then, by joining these arrows together to form trajectories, we can obtain a picture of how the populations will behave, as shown in Fig. 2.3.

We decide on the directions of the arrows by examining Equations 2.10. Thus

- (a) If $1 - bx - cy > 0$, then $\dot{x} > 0$, and the arrow must tend upwards: if $1 - bx - cy < 0$, the arrow tends downwards.
- (b) If $1 - fx - gy > 0$, then $\dot{y} > 0$, and the arrow must tend to the right: if $1 - fx - gy < 0$, the arrow tends to the left.

The trajectories in Fig. 2.3 are deduced from these facts. Provided that, initially, neither x nor y equals zero, the system will move to a stable equilibrium, with both types present.

Of course, it cannot be true that two types can always coexist, regardless of the nature of the interactions between them. Figure 2.3 assumes that $b > f$ and $g > c$. In words, this means that the inhibiting effect of each species on itself is greater than its inhibiting effect on the other. This would be true, for example, if the two types compete for some kinds of food, but if there are also kinds of food taken only by one type or by the other.

If one, or both, of these inequalities are reversed, stable coexistence is no longer possible. It is a useful exercise to draw the trajectories in state space when both inequalities are reversed (see Problem 6).

try to make this idea quantitative. How accurate must replication be if adaptation is to be maintained? This question is discussed in Box 2.3.

Equation 2.13 gives the critical value of Q , the accuracy of replication, if the

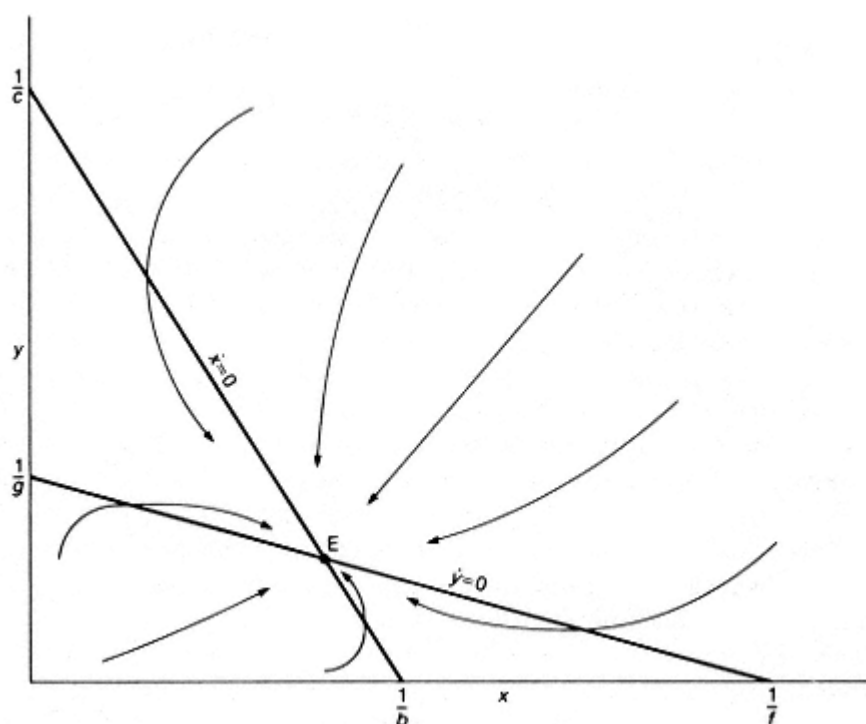


Figure 2.3

Trajectories in state space for competition between two types, with densities x and y , described by Equations 2.10. The bold lines are the loci of points for which $\dot{x} = 0$, and $\dot{y} = 0$. The point E is a stable equilibrium.

Box 2.3—

The Accuracy of Replication

Imagine a population of replicating RNA molecules. There is some unique sequence, S , that produces copies at a rate R : all other sequences produce copies at a lower rate, r . Thus sequence S is in some way superior to all others. A sequence produces an exact copy of itself with probability Q . If x_0 and x_1 are the numbers of copies of S and non- S respectively, then, ignoring deaths,

$$\begin{aligned} dx_0/dt &= RQx_0, \\ dx_1/dt &= rx_1 + R(1 - Q)x_0. \end{aligned} \quad (2.11)$$

In writing down these equations, I have assumed that when an error occurs in the replication of a non- S sequence, it gives rise to another non- S sequence: that is, I have ignored the very small probability that a non- S sequence will give rise to an S sequence.

We are interested in the survival of optimal molecules. At first sight, Equation 2.11 suggests that they will survive provided that $RQ > 0$. If there is

no mortality, this is correct: if an optimal molecule survives for ever, then the number of such molecules will increase provided that some copies, however few, are also optimal. But of course there will be some mortality. What, then, is the minimum value of Q , given mortality? Adding the two equations gives

$$d(x_0 + x_1)/dt = Rx_0 + rx_1 - D(x_0 + x_1).$$

Can optimal molecules survive when deaths balance births (as must be the case in the long run)? That is, when

$$D = \frac{Rx_0 + rx_1}{x_0 + x_1} = Rp + r(1 - p), \quad (2.12)$$

where p is the proportion of the molecules in the population that are optimal. The equation for x_0 , the number of optimal molecules, is

$$dx_0/dt = RQx_0 - \{Rp + r(1 - p)\}x_0.$$

Hence, at equilibrium,

$$Q = p + \frac{r}{R}(1 - p).$$

As the accuracy, Q , decreases, so the proportion, p , of optimal molecules at equilibrium will also decrease. The lowest value of Q compatible with any optimal molecules being present occurs when $p \simeq 0$, and

$$Q \simeq r/R. \quad (2.13)$$

adapted sequence, S , is to be maintained by selection against the deterioration caused by mutation. If the replication rate of the mutants is only slightly less than that of the optimal sequence (i.e. weak selection), then the accuracy Q must be high, because the mutant particles compete with S for resources. What if mutants replicate slowly: in the extreme case, suppose they do not replicate at all? It does not follow that any degree of accuracy, however low, will be sufficient. Thus S particles will not be immortal: there will be some rate of destruction, or 'death rate', even in the absence of competition from non- S particles. On average, each S particle must, during its life, produce one perfect S copy. Hence if the average number of copies per S particle before it is destroyed is R , then $Q > 1/R$ is necessary.

The critical accuracy, then, depends both on the success of non- S copies, and, if non- S particles have a low replication rate, on the average number of copies produced by an S particle during its lifetime. In practice, it seems unlikely that evolution would be possible if $Q < 1/2$.

The practical implication of this is that it places a limit on the size of the genome,

for any given replication accuracy. Thus consider a genome of n nucleotides, and let the probability that an error is made in replication be u per nucleotide. Then $Q = (1 - u)^n \cong e^{-nu}$. Hence the maintenance of adaptation requires, very approximately, that $nu < 1$. Three very different error rates exist: the rate for replication in the absence of enzymes, which may have occurred during the origin of life; the rate for replication of RNA, which does not involve a 'proof-reading' stage; and the rate for the replication of DNA, with proof-reading. The values are, very approximately, as follows:

	error rate (u)
non-enzymic replication	1/10-1/100
RNA replication	10^{-3} - 10^{-4}
DNA replication	10^{-9} - 10^{-10}

The requirement that $nu < 1$ then explains the fact that the genome of RNA viruses is never greater than about 10^4 bases, and of higher organisms no greater than 10^9 bases. It also raises an important difficulty for theories of the origin of life. The genome could not become greater than 100 bases in the absence of specific replication enzymes, yet a genome of less than 100 bases could hardly code for such an enzyme: for further discussion of this problem, see Eigen *et al.* (1981), and Maynard Smith and Szathmari (1995).

Genetic Drift in Finite Populations

The models considered so far have been deterministic: that is, they have assumed that the proportions of individuals of different kinds in the next generation is exactly what would be expected from the known probabilities of survival and reproduction. It is as if we were to assume that, if we toss a coin 100 times, we will get exactly 50 heads. This deterministic assumption, as far as proportions or frequencies are concerned, is approximately true if the population is large, and exact only if the population is infinite. In small populations it may be seriously in error. The proportions of different kinds will fluctuate by chance: such fluctuations are referred to as **genetic drift**.

To get some idea of the magnitude of this effect, suppose that we have an asexually reproducing population of N individuals, in each generation. There are two types, a and A , which do not differ in likelihood of survival or reproduction: that is, there is no selection. In one generation, let the frequencies of a and A be p and q , respectively: that is, there are Np of type a and Nq of type A . In reproduction, each type produces offspring like itself. Let p' and q' be the frequencies of a and A in the next generation. In an infinite population, $p' = p$ and $q' = q$. If the population is of finite size N , however, the frequencies in the next generation may not be the same as the frequencies in this, because some individuals will, by chance, have more offspring than others. It is helpful to imagine producing, not

one offspring generation of size N , but repeating the process many times. If we then measure the frequency, p' , of a in each trial, we can calculate the expected value of p' : that is, the average value in repeated trials. If there is no selection, the expected value is unchanged: that is $E(p) = p$ and $E(q) = q$. However, p' will vary from trial to trial, because of chance fluctuations. We can measure the variability of p' by its variance: that is

$$V = E(p' - \bar{p}')^2,$$

where $\bar{p}'$ is the mean value of p' . Since $\bar{p}' = p$, we have that $V = E(p' - p)^2$. Alternatively, we can measure the variability of p' by its standard deviation.

$$\sigma(p') = V^{1/2}.$$

The variability of p' will depend on the variability of family size. Thus if every member of the population has exactly one offspring, then $p' = p$ and $V = 0$. A more realistic assumption is that family size has a **Poisson distribution**: this is the distribution that would be obtained in a large population if each of the total of N offspring was assigned randomly to one of the N parents, independently of how the other offspring were assigned. If so, each offspring has a probability p of being a . The probabilities that there will be 0, 1, 2, . . . N offspring of type a are given by successive terms of the **binomial distribution**, $(p + q)^N$: that is

$$p^N, Np^{N-1}q, \frac{N(N-1)}{2} p^{N-2}q^2, \dots, q^N.$$

The binomial distribution has the properties shown in Table 2.1. Thus suppose that $p = q = 1/2$. Then

$$\sigma(p') = \sqrt{\frac{pq}{N}} = \frac{1}{2\sqrt{N}}.$$

For populations of various sizes:

Population size, N	10	100	1000	10^6
p' { mean	0.5	0.5	0.5	0.5
SD	0.158	0.05	0.0158	0.0005

Table 2.1

The binomial theorem. Type a individuals occur with probability p in a population of N .

The probabilities of 0, 1, 2 . . . N individuals of type a are given by the terms of $(p + q)^N$.

	Number of a individuals	Frequency of a individuals
Mean	Np	p
Variance	Npq	pq/N
Standard deviation	$\sqrt{Npq}$	$\sqrt{pq/N}$

Thus in a population of 100, if there are 50 a individuals in one generation, the number may well be less than 45, or more than 55, in the next: in fact, there is a chance of about one-quarter that the number will lie outside these limits. It is more difficult to estimate by how much the frequency is likely to have changed after, say, 10 to 100 generations, because the fluctuations will be sometimes upwards and sometimes downwards. The problem is treated in Box 2.4. The important points to bear in mind are as follows:

1. In a finite population, the frequencies of different types will fluctuate from generation to generation, in the absence of natural selection. Therefore, if one wishes to demonstrate that selection has occurred by comparing the relative frequencies of different types in successive generations, one must show that the changes are greater than those that would be expected by chance.
2. The smaller the population, the greater the fluctuations in frequency.
3. In a finite population, one type will ultimately become fixed by chance, all others having been eliminated. Figure 2.4 illustrates this process, for two initially equally frequent types, in a population of 50 individuals.

**Box 2.4—
Genetic Drift**

Imagine a population of N asexual individuals, with separate generations. The N individuals in the next generation are produced one by one, and each new individual is equally likely to be produced by any one of the N parents.

In some future generation, all the individuals will, by chance, be descended from a single individual in the present generation. How long will this take? There are two different ways in which we might ask this question, but they have the same answer:

1. How long will it be before the whole population is descended from one individual in the present generation?
2. How many generations must we go into the past to find the single common ancestor of the whole present population?

The answer is that the expected time is $2N$ generations, with a standard error a little greater than N . This conclusion holds for any constant population of N replicators, if the number of copies of a particular replicator has a Poisson distribution. It is applied to the spread of a selectively neutral gene, and to mitochondria on p. 153. The result is derived by a method, the diffusion approximation, that is beyond the scope of this book. It is well explained by Roughgarden (1979). For sceptics, it is easy to check the result by simulation.

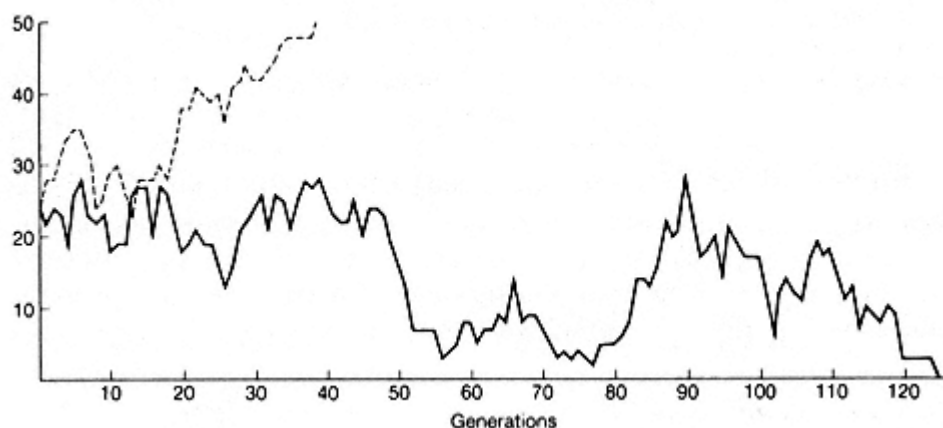


Figure 2.4

Genetic drift. Two selectively equivalent types, a and A , reproduce asexually, in a total population of $N = 50$. Family size is Poisson-distributed. The graphs show the number of a individuals in two of the first three simulations carried out: the third simulation lasted 227 generations before all N individuals were a .

Further Reading

Eigen, M., Gardiner, W., Schuster, P., and Winkler-Oswatitsch, R. (1981). The origin of genetic information. *Scientific American* **244**, 88-118.

Maynard Smith, J. (1974). *Models in ecology*. Cambridge University Press.

Problems

1. A population of size x has a constant birth rate, r , and a death rate of cx : that is, an individual has a probability of dying that is proportional to the population size. Write down a differential equation describing the growth of the population. What is the carrying capacity?
2. In a population of RNA molecules 100 nucleotides long, the error rate for a cycle of replication (A-U-A or G-C-G) is 0.001 per nucleotide. (a) What is the probability of the exact replication of the whole molecule? (b) A unique sequence, S , replicates at a rate 20 per cent greater than all other sequences, which have equal rates. At equilibrium between selection and mutation, what fraction of the population consists of S sequences? (c) What is the maximum error rate per nucleotide compatible with the existence of S at equilibrium?
3. Suppose that you have the numbers from which Fig. 2.4 was plotted, and you suspect that the numbers were invented, and are not the result of simulation. Suggest some simple statistical tests that might confirm your suspicion.
4. In an asexual population of 1000, there are 700 true-breeding individuals of kind A , and 300 of kind B . Assuming no selection, what, approximately, is the probability that there will be more than 73 A individuals in the next generation? What have you assumed?
5. In a diploid population of 500, how many generations will elapse, on average, before all

the alleles at a selectively neutral locus are replicas, with or without mutation, of a single allele in this generation?

6. If, in the model described in Box 2.2, $b < f$ and $g < c$, what will happen?

Computer Projects.

Simulate genetic drift. Start with a population of N asexual types, all different. Produce a new population by random sampling, and continue until all are identical. Use the program to check the statement that the expected time is $2N$ generations, with a standard deviation a little greater than N .

Chapter 3— Evolution in Diploid Populations

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In this chapter, I consider for the first time a sexually reproducing population. I shall consider diploids, because most sexual species are diploids, but it is well to remember that many algae are both haploid and sexual. I shall also assume that both population sizes and fitness differences are large enough to justify a deterministic treatment: chance events are discussed in Chapter 8.

Gene Frequencies and the Hardy-Weinberg Ratio

This chapter is concerned with changes in the frequencies of genes in large populations. I start by defining the 'frequency' of a gene. Consider a large population of sexually reproducing diploids. Suppose that at some autosomal locus there are two possible alleles A and a (I use upper- and lower-case symbols so as to avoid unnecessary subscripts, and not to imply dominance and recessivity). There are then three possible genotypes, AA , Aa , and aa . Imagine that we count a large sample, N , of the population, classifying each one according to its genotype, with the following results:

Genotype	AA	Aa	aa	Total
Number	n_1	n_2	n_3	N
Frequency	$P = n_1/N$	$Q = n_2/N$	$R = n_3/N$	

From the numbers, we can calculate the genotype frequencies, P , Q , and R , as shown. Note that $P + Q + R = 1$.

We now define the frequency of the allele A as the number of A genes in the population, divided by the total number of genes: in performing this calculation, we treat each individual as if it contained exactly two genes. Thus, writing p for the frequency of A , and $q = 1 - p$ for the frequency of a , we have:

$$p = (2n_1 + n_2)/2N; q = (n_2 + 2n_3)/2N, \quad (3.1)$$

or, equivalently, $p = P + 1/2Q$; $q = 1/2Q + R$; and, obviously, $p + q = P + Q + R = 1$ (see Fig. 3.1).

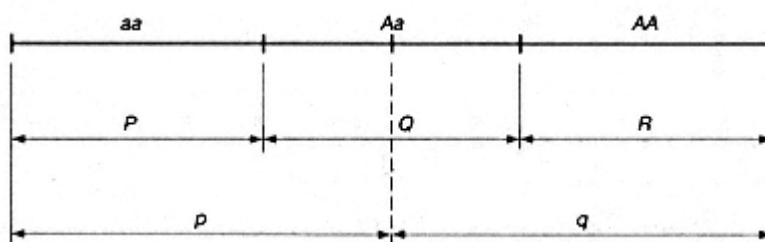


Figure 3.1
The relationship between the genotype frequencies at a locus,
 P , Q , and R , and the gene frequencies p and q .

This definition can easily be extended to any number of alleles at a locus. If we know the numbers, or frequencies, of all the possible genotypes, it is a trivial matter to calculate the frequencies of the alleles. Can we do the reverse? That is, can we calculate P , Q , and R when we know p and q ? In general, we cannot: there is no way of calculating three unknowns from two equations. However, we can do so if we assume that mating is 'random': that is, if the genotypes (at this particular locus) of the two parents of a child are independent. There are two ways of tackling the problem: they give the same answer, but it is worth following both.

The Gene Pool Approach

Imagine that we look at a random zygote (i.e. a fertilized egg) just after fertilization. It will have received one allele, A or a , from its father, and one from its mother, according to the following scheme:

gene from father	gene from mother	joint probability
A	A	$p \times p = p^2$
A	a	$p \times q$
a	A	$q \times p$
a	a	$q \times q = q^2$

In this table, it is assumed that the joint probability of getting, say, A from father and a from mother is the product of the two separate probabilities. This is justified only if the two events are independent of one another, and hence only if mating is random. Thus suppose, as a counter-example, that allele a is for blue eyes (so that aa genotypes have blue eyes, and Aa and AA have brown eyes), and also that there is assortative mating for eye colour, so that blue-eyed people tend to marry one another. Then if a child gets the gene a from its father, it is somewhat more likely to get a from its mother also than would be the case if it got gene A from its father.

The conclusion, then, is that under random mating we expect the three genotypes in the proportion p^2 AA : $2pq$ Aa : q^2 aa , Where p and q are the frequencies of alleles A and a among the gametes. This is the famous 'Hardy-Weinberg' ratio, named after its two independent discoverers. Note that it is achieved after one

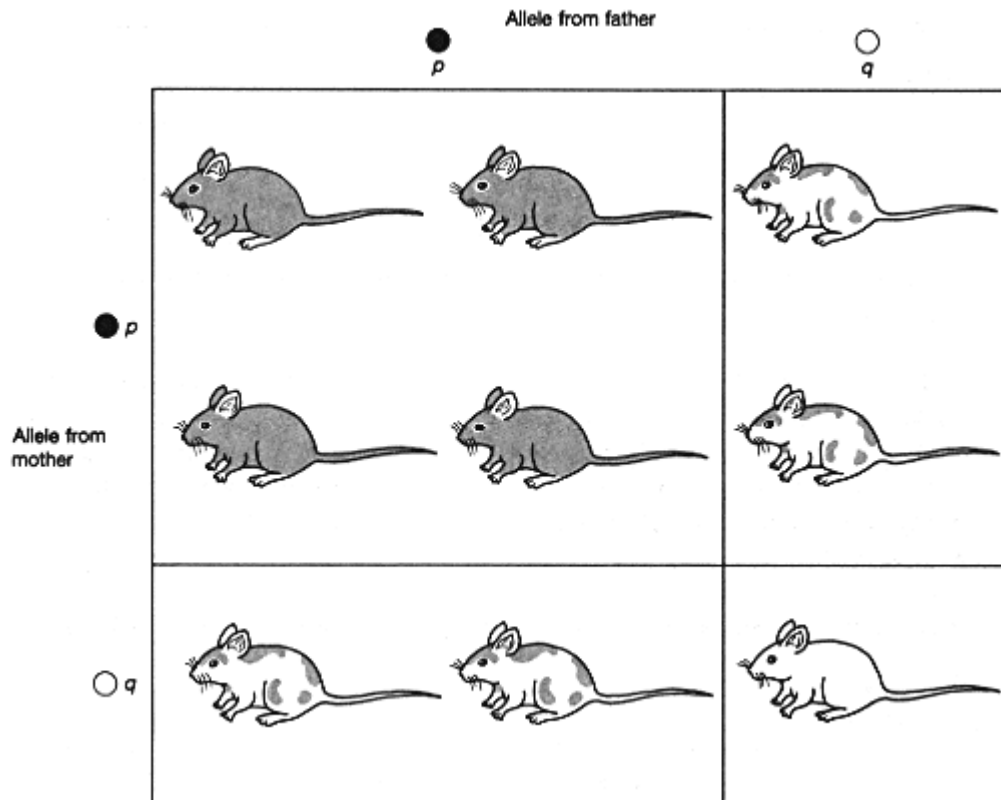


Figure 3.2
A geometrical representation of the Hardy-Weinberg ratio.

generation of random mating: the offspring will be in the Hardy-Weinberg ratio even if the parents were not. A geometrical representation of the argument is given in Fig. 3.2.

It is important to note several assumptions that were made in deriving the result:

1. Mating is random.
2. The frequencies p and q are the same in males and females.
3. p and q are the frequencies among successful gametes: that is, gametes that contribute to zygotes. Hence it is not enough to calculate p and q from the frequencies P , Q , and R of the genotypes in the adult population. These genotype frequencies must be weighted by their fertilities.
4. The frequencies among zygotes are recorded before any selection has acted.

I headed this section the **gene pool** approach because it is equivalent to supposing that the adult population contributes genes to a gene pool, and that new diploid zygotes are formed by drawing two genes, at random, from this pool. We

shall meet this way of thinking again in Chapter 8. I now turn to an alternative way of deriving the Hardy-Weinberg ratio.

The Mating Table Approach

As before, let the frequencies of genotypes AA , Aa , and aa in the adult breeding population be P , Q , and R . Table 3.1 lists the different possible matings, their frequencies, and the offspring produced. Note that the frequency of the mating $AA \times Aa$ is $2PQ$, because there are two possible cases, $AA \text{ ♀} \times Aa \text{ ♂}$, and $Aa \text{ ♀} \times AA \text{ ♂}$, each with frequency PQ . Adding up the offspring gives $p^2 AA : 2pq Aa : q^2 aa$. Note that the four assumptions listed above (random mating, equal gene frequencies in the two sexes, breeding adults and not total adults counted, no selection) have again been made.

In deriving the Hardy-Weinberg ratio, it is easier to use the gene pool approach, and there is no real need to set out the full mating table. However, it is useful to introduce mating tables here, because they are needed in more complex cases or example, if mating is not random.

If a sample is found not to be in the Hardy-Weinberg ratio, there are various possible explanations:

1. Selection has operated between the formation of zygotes and the time of sampling.
2. Mating is not random. This may be caused by the alleles themselves: for example, if alleles in a plant affect flowering time, mating for those alleles would not be random. It can also be for reasons that have nothing to do with the alleles themselves: for example, if mating between relatives is common, then mating will not be random even for alleles which have no effect on mating.

Table 3.1

The Hardy-Weinberg ratio derived from a mating table.

Mating	Frequency	Progeny		
		AA	Aa	aa
$AA \times AA$	p^2	p^2	—	—
$AA \times Aa$	$2PQ$	PQ	PQ	—
$AA \times aa$	$2PR$	—	$2PR$	—
$Aa \times Aa$	Q^2	$Q^2/4$	$Q^2/2$	$Q^2/4$
$Aa \times aa$	$2QR$	—	QR	QR
$aa \times aa$	R^2	—	—	R^2
Totals	$(P+Q+R)^2$	$(P+Q/2)^2$	$2(P+Q/2)(Q/2+R)$	$(Q/2+R)^2$
	$=1$	$=p^2$	$=2pq$	$=q^2$

3. The sample is not from a single population. It is shown in Box 3.1 that if a sample is composed of two sub-samples, each from a random-mating population, but with different frequencies of allele A , then in the total sample there will be more homozygotes and fewer heterozygotes than would be predicted by the Hardy-Weinberg ratio. This, the **Wahlund effect**, is a common cause of the departure of real data from the Hardy-Weinberg ratio.

4. Genotypes have different likelihoods of being included in the sample.

As an application of the Hardy-Weinberg ratio, consider the data collected by da Cunha on *Drosophila polymorpha*. Two alleles at a locus determine abdomen colour: EE is dark, Ee intermediate, and ee is pale. Table 3.2 gives the frequencies of these types among 8070 flies collected in Brazil. There is a highly significant deficiency of heterozygotes, compared to the numbers expected from the Hardy-Weinberg ratio.

One possible explanation is that heterozygotes are less viable in the wild, but this is unlikely, because laboratory measurements suggested that heterozygotes had a higher viability than either homozygote. It is conceivable that abdomen colour influences mate choice: if there is a tendency for like to mate with like, this could explain the discrepancy. Perhaps the most plausible explanation is that the flies are not drawn from a single random-mating population. It was shown in Box 3.1 that if a total sample is made up of several sub-samples, each of Hardy-

**Box 3.1—
The Wahlund Effect**

Suppose that two alleles are segregating in a species. Two samples, each of size $N/2$, are taken from different random-mating populations, with frequencies of A of 0.2 and 0.7, respectively. If the two samples are pooled, the expected numbers in the joint sample of N are:

$$AA: (0.04 + 0.49) N/2 = 0.265N$$

$$Aa: (0.32 + 0.42) N/2 = 0.370N$$

$$aa: (0.64 + 0.09) N/2 = 0.365N$$

$$\text{Total} = N$$

Thus the overall frequency of A is $(2 \times 0.265 + 0.37)/2 = 0.45$, and hence, if the pooled sample had been taken from a single random-mating population, the expected frequencies of the genotypes would be 0.2025 AA :0.495 Aa :0.3025 aa . The actual numbers in the pooled sample show a deficiency of heterozygotes relative to these expectations.

It is not difficult to show that such a deficiency of heterozygotes is to be expected whenever a total sample is made up by pooling samples from random-mating populations with different gene frequencies.

Table 3.2

Abdomen colour in *Drosophila polymorpha* (Cunha 1949).

	Gentotype dark EE	intermediate Ee	light ee	Total
Observed (<i>O</i>)	3969	3174	927	8070
Expected from H-W ratio (<i>E</i>)	3825	3462	783	
<i>O-E</i>	144	-288	144	
$(O-E)^2/E$	5.42	23.96	26.48	55.86

$$p(E) = \frac{2 \cdot 3969 + 3174}{2 \cdot 8070} = 0.6885$$

$$p(e) = 1 - p(E) = 0.3115$$

χ^2 with one degree of freedom = 55.86. $P < 0.001$. (Note that there is only one degree of freedom, because p has been calculated from the data.)

Weinberg ratio but with different gene frequencies, then the total sample will have an excess of homozygotes.

Perhaps the main utility of the Hardy-Weinberg ratio is in developing theoretical models: if, in a model, we assume random mating, we reduce the number of variables with which we have to deal. However, it does have some empirical value in providing a null hypothesis when seeking evidence of natural selection or of assortative mating. The snag is that it is not a particularly sensitive way of detecting selection (see Problem 3). Also, it will fail to detect selection arising from differences of fertility rather than viability.

The Concept of Fitness

Before we can calculate how the frequencies of genes in a population will change in time, we need some measure of the survival and reproduction of the different genotypes. This measure is **fitness**, which has a technical meaning in population genetics. The definition is easiest for populations with separate generations. For example, an annual insect in which all adults that breed in one year die before the next. Then the fitness of a particular type, A , is the expected number of offspring contributed by an A individual to the next generation. Fitness is estimated from one particular stage in the life cycle. Usually the zygote to the corresponding stage in the next (see Fig. 3.3). Then the fitness of zygotes of type A is the expected number of zygotes in the next generation to which an A zygote in this generation will contribute gametes. That is:

'fitness of an A zygote' = 'probability that an A zygote will survive to breed' × 'expected number of offspring, given that it does survive'.

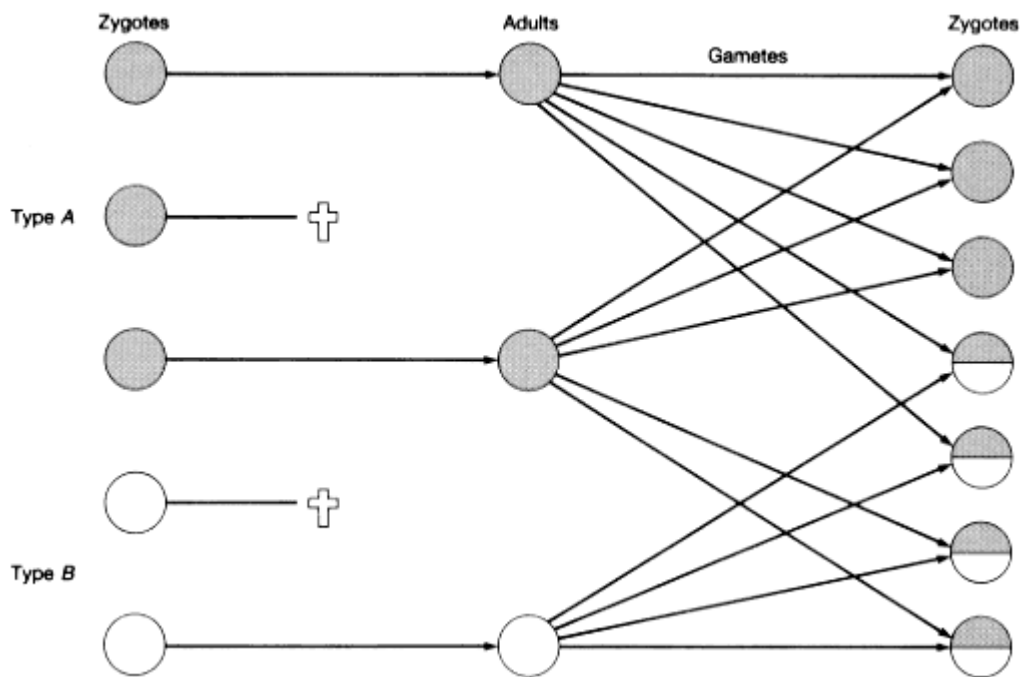


Figure 3.3

The definition of fitness when generations are separate. Type A has a $2/3$ chance of surviving, and survivors contribute five gametes to the next generation, giving an absolute fitness of $2/3 \times 5 = 10/3$. Type B has a $1/2$ chance of surviving, and contributes four gametes, giving an absolute fitness of 2. If we are concerned only with the relative fitnesses of different zygotes, it would be convenient to take the fitnesses of A and B as $W(A) = 1$, and $W(B) = 2 \div 10/3 = 0.6$. Note that the diagrams represent average values of survival and fertility: individuals would vary either side of those averages. Note also that, in defining fitness, we are concerned only with the performance of genotypes: we do not keep track of particular alleles.

What has just been defined is 'absolute' fitness. More often, however, population geneticists use 'relative' fitnesses. For example, a model may assume that the fitnesses of genotypes AA , Aa , and aa are 1, 1, and 0.5 respectively. If these were absolute fitnesses—that is, actual numbers of offspring—then the population size would decline in every generation. In practice we make the (often tacit) assumption that population numbers are regulated, and that all we are concerned with is the change in gene frequency in a constant population. It is therefore sufficient to use relative fitnesses, reducing one of the fitnesses to unity, and scaling the others by the same factor. For example:

Genotype	AA	Aa	aa
Absolute fitness	W_1	W_2	W_3
Relative fitness	1	W_2/W_1	W_3/W_1

Some assumptions lying behind the definition of fitness need clarification:

1. Fitness is a property, not of an individual, but of a class of individuals. For example, of individuals homozygous for allele A at a particular locus. Thus the phrase 'expected number of offspring' means the average number, not the number produced by some one individual. If the first human infant with a gene for levitation were struck by lightning in its pram, this would not prove the new genotype to have low fitness, but only that the particular child was unlucky. Usually we ascribe a fitness to a 'genotype', meaning a class of individuals with some genetic characteristic in common.
2. Fitness is specific to an environment. Thus in my lifetime, myopic individuals have had a high fitness, because they were less likely to be put in military uniform and shot at. In a hunter-gatherer society, myopics would probably have a low fitness. We can, if we wish, define fitness not for one specific environment, but for a set of environments, each encountered with some probability.
3. Fitness is measured over one generation. It requires further computation to predict what will happen over many generations.
4. Fitness is a property of classes of individuals, and not genes. The reason for this is that selection typically acts on individuals. However, we will meet contexts in which it is useful to ascribe fitnesses to so-called 'marginal fitnesses' of genes: an example occurs on p. 65.
5. Populations do not have fitnesses. Essentially, this is because populations do not reproduce, and hence there is no 'generation' over which fitness could be measured. Possible exceptions are discussed in Chapter 9 (the units are populations of viruses in a single host), and in Chapter 14 (the units are species).

Complications arise if generations are not separate. Thus, consider a species like our own. Let there be two genotypes, one of which has an average of three children before the age of 25, and none after, and the other of which has four children after age 25, and none before. Which is fitter? And by how much? R.A. Fisher proposed that, in such cases, fitness is best measured by the 'Malthusian parameter', defined in Box 3.2. Essentially, this parameter is the intrinsic rate of increase of the genotype in question. Fortunately, most conclusions based on models with separate generations extend to the more complex case with overlapping ones. We need to bother about these complications only if we are interested in the evolution of agespecific traits. For example, senescence, or age of first breeding.

The Spread of a Favourable Gene

Consider the spread of a favourable gene in a random-mating population. Let A , a be the favourable and unfavourable alleles, respectively, and let p_n be the frequency of A among new zygotes in generation n . The fitness of AA is $1 + s$, of

Box 3.2— Overlapping Generations

It is hard to define fitness when generations overlap, as they do in humans. As a start, we need a description of the growth of a population of identical individuals. For simplicity, suppose that a female reproduces at a series of discrete moments—say at age 1, 2, 3 . . . years, and so on (for continuous breeding, the summation signs below would have to be replaced by integral signs).

Let ℓ_x = probability that a female survives to age x , and m_x = expected number of female offspring produced by a female of age x .

Note that males are relevant only in ensuring that females are fertilized. Then the total expected female offspring to a female in her lifetime is

$$R_0 = \sum_{x=0}^{\infty} \ell_x m_x. \quad (3.2)$$

R_0 is called the net reproductive rate. If $R_0 > 1$ the population will grow. There is an important theorem, due to Lotka, which says that, if the values of ℓ_x and m_x remain constant, then after a sufficient period of time the population will acquire a 'stable age distribution'. That is, the proportion of females of any age—say 30 years—will remain constant. The total population will then increase exponentially at a rate r ; that is, $N = N_0 e^{rt}$, and so will the number of females of any particular age. To find r , we solve the equation

$$\sum_{x=0}^{\infty} \ell_x m_x e^{-rx} = 1. \quad (3.3)$$

The simplest way to solve this equation is by trial and error, substituting various values of r .

Why should this equation be true? To see this, imagine a population that has reached its stable age distribution. Let the number of females born in some particular year, t_0 , be T . Then, of these T females, the number born to females of age n is equal to (number of females born n years ago) $\times \ell_n \times m_n$. But the number of females born n years ago is $T e^{-rn}$. Hence, summing over all values of n , $\sum \ell_n m_n T e^{-rn} = T$, which is Equation 3.3.

r is called the intrinsic rate of increase, or Malthusian parameter. Suppose now that the population consists of genetically different females, with different values of ℓ_x and m_x . R. A. Fisher proposed that the appropriate measure of fitness of a genotype in a sexual population is the corresponding value of r . The full justification of this proposal is beyond the scope of this book. However, some feel for it can be obtained by considering a population consisting of a mixture of asexually reproducing females, each with characteristic values of ℓ_x and m_x , and hence of r . Once each type had reached a stable age distribution, each would be increasing at a rate measured by its Malthusian parameter: the type with the largest value of the parameter would increase in frequency relative to the others.

Some numerical cases will help to illustrate how this works out:

Case 1

Type A females all survive for 5 years, and then produce 6 female offspring and die. Then $R_0 = 6$, and $6e^{-5r} = 1$, or $r = 0.358$.

Type B females all survive for 2 years, and then produce 3 female offspring and die. Then $R_0 = 3$, and $3e^{-2r} = 1$, or $r = 0.549$.

In this case, Type B is fitter than Type A, although it has a lower net reproductive rate. In a growing population, it pays to reproduce young.

Case 2

As case 1, but each type of female produces only one female offspring.

For Type A, $R_0 = 1$, and $e^{-5r} = 1$, or $r = 0$.

For Type B, $R_0 = 1$, and $e^{-2r} = 1$, or $r = 0$.

The two types are of equal fitness. In a stationary population, the genotype with the higher value of R_0 has the higher fitness.

Case 3

As case 1, except that both types of female produce only 0.5 female offspring.

For both types, $R_0 = 0.5$.

For Type A, $0.5e^{-5r} = 1$, or $r = -0.139$.

For Type B, $0.5e^{-2r} = 1$, or $r = -0.346$.

Type A is fitter. In a diminishing population, it pays to breed late.

The weakness of this model of growth and selection in populations with overlapping generations is that there is no mechanism of population regulation. In real populations, either ℓ_x or m_x would decrease with increasing population size.

Aa is $1 + hs$, and of aa is 1. By choosing the value of h we can consider a dominant favourable allele ($h = 1$), a recessive allele ($h = 0$), or an allele of intermediate dominance ($h = 1/2$). For the moment, we will assume that the fitness differences arise because different genotypes have different probabilities of survival (viabilities), and that, if they do survive, the three types are equally fertile. Then, in generation n

Genotype	AA	Aa	aa
Fitness	$1 + s$	$1 + hs$	1
Frequency of zygotes	p_n^2	$2p_nq_n$	q_n^2
Relative proportions of adults	$p_n^2(1 + s)$	$2p_nq_n(1 + hs)$	q_n^2

I have written 'relative proportions' rather than 'frequency' in the last row because the entries do not add up to one. Thus the sum of the three entries is:

$$\begin{aligned}
 T &= p_n^2 + sp_n^2 + 2p_nq_n + 2hsp_nq_n + q_n^2 \\
 &= 1 + s(p_n^2 + 2hp_nq_n).
 \end{aligned}
 \tag{3.4}$$

The frequency of allele A in the next generation of zygotes, p_{n+1} , is equal to its frequency among the adults of generation n . That is:

$$p_{n+1} = [2 \times \text{number of } AA \text{ adults} + \text{number of } Aa \text{ adults}] / 2T,$$

or

$$p_{n+1} = \frac{p_n^2(1+s) + p_nq_n(1+hs)}{1 + s(p_n^2 + 2hp_nq_n)}. \quad (3.5)$$

Suppose that, instead of having different viabilities, the three genotypes were equally likely to survive to breed, but that their fertilities were in the ratio $1 : s : 1 + hs : 1$. Then the last row of the table no longer gives the relative proportions of adults (which are $p^2 : 2pq : q^2$), but it does give the relative contributions of AA, Aa, and aa adults to the gene pool from which the next generation is formed. Hence Equation 3.5 still holds.

What we would like to get from this equation is an expression for p_n as a function of the initial frequency, p_0 , and the fitness coefficients s and h . Unfortunately, we cannot in general do that. We can, however, proceed in one of two ways. First, we can look at the initial increase in A when it is rare (p small): this is done in Box 3.3. It turns out that if A is completely recessive ($h = 0$), the initial increase is very slow. If A is partially or wholly dominant, p_n is given by

$$p_n = p_0 e^{shn}. \quad (3.6)$$

Box 3.3—

The Increase in a Favourable Gene When Rare

Equations such as 3.5 are called finite difference equations. Provided that the change in one time interval is small, such an equation can be approximated by a differential equation. To do this, we find an expression for δp_n , the change in p_n in one generation. That is

$$\delta p_n = p_{n+1} - p_n = \frac{p_n^2(1+s) + p_nq_n(1+hs)}{1 + s(p_n^2 + 2hp_nq_n)} - p_n.$$

After some fiddling, this reduces to

$$\delta p_n = \frac{sp_nq_n[p_n + h(1 - 2p_n)]}{1 + s[p_n^2 + 2hp_nq_n]}.$$

If s is small, the change in p in a single generation is small, and we can replace this by a differential equation. Since, for s small, the denominator is approximately equal to 1, we have

$$\frac{dp}{dt} = spq[p + h(1 - 2p)]$$

where t is measured in generations.

If we are interested only in the initial increase of A (p small, $q \approx 1$), this becomes

$$\frac{dp}{dt} = sp(p + h).$$

If A is partially or wholly dominant, we have

$$\frac{dp}{dt} = shp, \quad \text{or} \quad p_n = p_0 e^{shn}.$$

If A is recessive ($h = 0$), then $dp/dt = sp^2$, and for small p the rate of increase is very slow.

That is, A increases exponentially, at a rate determined by the selective advantage sh , in the heterozygotes. These results make intuitive sense. If A is rare, it occurs primarily in heterozygotes, so that its rate of increase will depend on sh . If it is wholly recessive, it will be shielded from selection, and so will increase very slowly.

Alternatively, we can easily solve Equation 3.5 on a computer for any particular values of p_0 , h , and s . Figure 3.4 shows the results for $s = 0.1$ (a 10 per cent selective advantage), for the frequency change from $p = 0.01$ to $p = 0.99$. This confirms that the initial increase of a recessive gene is very slow. Note that the final fixation of the dominant gene is also very slow, for the same reason: when a recessive allele is rare it is not exposed to selection.

The classic example of the spread of a dominant gene under selection is the evolution of industrial melanism in the moth, *Biston betularia*. Prior to 1850, the only form of this moth recorded was of a speckled grey colour—the so-called 'typical' form. In 1850, an almost black moth, the '*carbonaria*' form, was reported from Manchester. A century later, up to 95 per cent of moths from industrial areas in Britain were *carbonaria*, although populations from rural areas are still mainly of the typical form. The black colour is caused by a dominant gene. A similar spread of melanic forms has been recorded from no less than 70 species of moth, all of which have the habit of resting during the day on exposed surfaces, in particular the trunks and branches of trees. In all but one of the analysed cases, the melanism that has been established is caused by partly or wholly dominant genes, although recessive mutations causing melanism are known to occur in several species.

An obvious explanation is that the change has been caused by selective predation. In industrial areas, the lichens on trees are killed by pollution, so that the background is dark, and the *carbonaria* form is hard to detect against it, whereas the typical form is hard to see on the lichen-covered trees in rural areas. Supporting this explanation is the fact that many bird species do feed on these

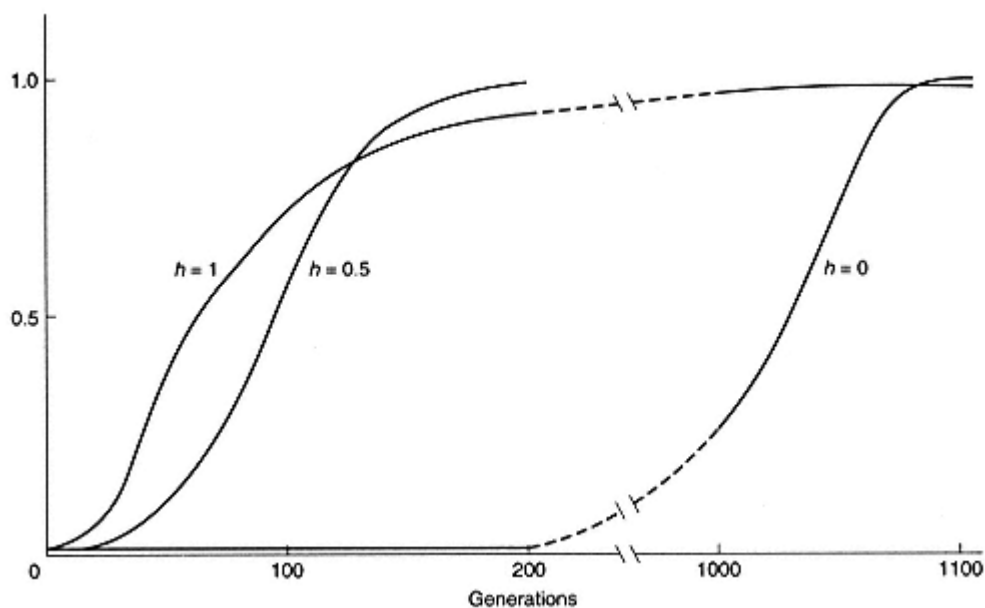


Figure 3.4
 Increase in frequency of a favourable gene. Fitnesses, $aa = 1.0$; $Aa = 1 + 0.1h$; $AA = 1.1$. Initial frequency of $A = 0.01$.

moths, and do find the typical form more easily on a polluted background. More direct evidence comes from mark-release experiments. Kettlewell marked moths with paint under the wings so that they could be recognized, and released them in an industrial area near Birmingham. Moths were later recaptured in a mercury-vapour light trap, with the results shown in the upper half of Table 3.3. The results are consistent with the selective hypothesis: a smaller proportion of the typicals were recaptured, suggesting that more had succumbed to predation. However, the data would also be explained if typical forms are more likely to move away from the site of release and recapture, or if they were less easily attracted to light traps. These alternatives are ruled out by the second experiment shown in the table: in a rural area in Dorset, a larger proportion of typicals were recaptured.

Why, in almost all cases, is the melanic gene that has spread dominant, or partly so? It is important to understand that this is not because only dominant mutations occur, or because only they are selectively favoured. It is because favourable recessive mutations spread very slowly. If, in some species, both dominant and recessive melanic mutations had been present in low frequency in 1800, only the dominant one would have had time to become common. Mutations with partial dominance spread almost as rapidly as complete dominants.

Of greater practical importance is the rapid spread of genetic resistance to insecticides (reviewed by Wood, 1981). The first case was recorded in 1908. Resistance had been reported in 14 species by 1948, in 224 by 1967, and in 364 by

Table 3.3

The numbers of moths, *Biston betularia*, released and recaptured in two regions (Kettlewell 1955).

	<i>carbonaria</i>	Typical	Total
Birmingham (urban)			
Released	154	64	218
Recaptured	82	16	98
% recaptured	53.2	25	
Dorset (rural)			
Released	473	496	969
Recaptured	30	62	92
% recaptured	6.3	12.5	

1976: the numbers continue to rise. On analysis, resistance in natural populations has always turned out to be due to one or a few genes. It is rarely wholly recessive: most commonly, the resistant allele is partially dominant.

There was at one time considerable argument about whether the relevant mutations were caused by the insecticides themselves. Since some insecticides are mildly mutagenic, one cannot rule this out, but it has been shown that the effect is not a necessary one. Bennett (1960) produced a strain of *Drosophila melanogaster* that was resistant to DDT, but whose ancestors had never been exposed to the insecticide. He did this by the process of family selection, illustrated in Fig. 3.5. In each generation, the offspring of each pair were divided into two groups, of which

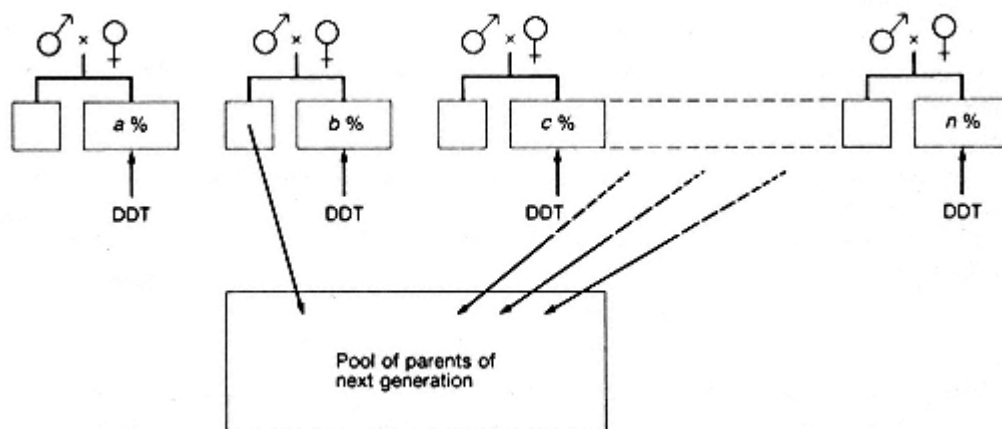


Figure 3.5

Design of sib-selection experiment for DDT resistance. Each family is split into two, and half is exposed to insecticide. The proportions surviving exposure are $a\%$, $b\%$, etc. Those families with the highest proportion of survivors (in the figure, it is assumed that $b\%$ is a high proportion) are used as parents of the next generation, but the parents are chosen from the unexposed half of the population.

one was exposed to DDT. The siblings of those groups with the highest resistance were then used as parents of the next generation. This sib-selection technique is a powerful way of demonstrating the non-Lamarckian nature of genetic adaptation.

How does genetic resistance work? In rough order of importance, the main mechanisms are:

1. *Detoxification*. For example, DDT is broken down by an enzyme (DDT-dehydrochlorinase) in *Musca*, *Aedes*, and other insects. DDT may also be oxidized.
2. *Alteration of the site of action*. For example, organophosphates work by inhibiting the enzyme AChE (acetylcholinesterase). Resistance has evolved by altering the AChE so that it is no longer inhibited in mites, ticks, blowflies, and mosquitoes, among others.
3. *Reduced penetration through the cuticle*. This is unusual, but a gene for impermeability has been established in houseflies.
4. *Behavioural avoidance of the insecticide*.
5. *Accelerated excretion*.

Further Reading.

The material on population genetics, in Chapters 3-6 and 8, is covered in the following textbooks, sometimes in greater detail:

Crow, J.F. (1986). *Basic concepts in population, quantitative and evolutionary genetics* W.H. Freeman, New York.

Hartl, D.L. (1980) *Principles of population genetics*. Sinauer, Sunderland.

Roughgarden, J. (1979). *Theory of population genetics and evolutionary ecology: an introduction*. MacMillan, New York.

A full account of the mathematical theory is given in:

Crow, J.F. and Kimura, M. (1970). *An introduction to population genetics theory*. Harper and Row, New York.

Problems

1. 500 mice were trapped on a farm and classified for the fast (F) and slow (S) electrophoretically detectable alleles at a locus, with the following results:

Genotype	FF	FS	SS	Total
Number	91	208	201	500

Are these results consistent with the Hardy-Weinberg ratio? If not, what explanations are there for the deviation? What observational tests of these explanations can you suggest? (Note: these data were invented.)

2. In London, 5 per cent of all male cats are ginger. What is the expected frequency of ginger females and of tortoiseshell females? (Ginger is a sex-linked recessive; tortoiseshell is the heterozygote.) What assumptions have you made?

3. Two alleles are present in equal frequencies in a random-mating population. The heterozygote has a viability 5 per cent greater than either homozygote. Would you expect to detect this difference in a sample of 1000 adults?
4. If mating is random, but the gene frequencies at a locus are different in breeding males and females, prove that there is an excess of heterozygotes, relative to the Hardy-Weinberg ratio, among their progeny.
5. At the ABO locus, three alleles determine four phenotypes according to the following scheme: genotypes AA and AO specify the A blood group; BB and BO specify the B group; AB specifies the AB group; OO specifies the O group. In a random mating population, the gene frequencies are $0.3A:0.2B:0.5O$. (a) What blood group frequencies are expected? (b) A child with blood group A has an A father and a B mother. If the child has a full sib, what is the probability that the sib is group A?
6. A monocarpic (flowering only once) species of plant is dioecious (separate sexes), with a 1:1 sex ratio. There are two types of female: type A produces 200 seeds at age 5 years, and type B produces 300 seeds at age 7 years. 98 per cent of seeds fail to give rise to plants; there is no other mortality before flowering. Calculate the net reproductive rate, R_0 , and the Malthusian parameter, r , for each type. Is the population increasing or decreasing? Which type has the higher fitness?
7. A gene with a frequency of $1/100\,000$ in a very large random-mating population increases fitness by 1 per cent. How many generations would it take to raise the gene frequency to $1/100$ if (a) the gene is dominant, (b) the gene is recessive? (Remember that, when p is small, $dp/dt = sp^2$; why might your answer to this part of the question be seriously misleading in practice?)

Computer Projects

Consider a population of females, living for a number of years, but breeding only once a year. For each age, choose a constant probability of surviving to that age, and a constant fertility for females that do survive. Write a program that will follow the age distribution in such a population, and use it to check that there is a stable age distribution. (If you know some matrix algebra, it would be interesting to look up the 'Leslie matrix', as a way of representing such a population, and to compare the idea of a stable age distribution with that of an eigenvector.)

Chapter 4— The Variability of Natural Populations

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The Evidence for Genetic Variability

There are two rather different ways in which a population may respond genetically to a changed environment. One, suggested by the examples of melanism and insecticide resistance, is as follows. When the new conditions first arise, the required mutations are either absent or very rare. There is an inevitable delay in the response of the population while the required mutations occur and increase from their initial low frequencies. The implication of Equation 3.6 is that, even for a fully dominant gene, it takes three times as long for the frequency to increase from, say, 1 in 100 000 to 1 in 100 as it does from 1 in 100 to 1 in 10: the subsequent increase from 1 in 10 will be rapid. During the first period, little observable change in the population would occur.

A rather different picture emerges from observations on one of Darwin's Galapagos Island finches, *Geospiza fortis*, on Daphne Major, a small islet of 40 hectares (Boag and Grant 1981). A serious drought in 1977 caused an 85 per cent decline in the population. At the same time, there was an increase in linear dimensions (bill, wing, tarsus) of from 2 to 5 per cent, because the larger birds were better able to survive on the large seeds that were the main available food during the drought. There is good evidence that differences in metrical traits in these finches are mainly genetic. Clearly this response did not depend on initially rare mutations. The relevant genetic variability must already have been present in the population. There are two main sources of information about genetic variability in natural populations. The first comes from the response to artificial selection. If a sample of a sexually reproducing species is brought into the laboratory, it will be found to respond to artificial selection for almost any trait. Box 4.1 lists some of the traits which have responded to selection in *Drosophila*, and also two traits for which no genetic variance was found. There is no reason to think that *Drosophila* is peculiar in this respect.

Given that populations respond to artificial selection in this way, how do we know that the response is due to genetic variance already present, and not to new

Box 4.1—***The Response of Drosophila Populations to Selection***

Characteristics that have been altered by artificial selection in *Drosophila* include the following: abdominal bristle number, wing length, longevity, ovary size, mating speed, tendency to move towards light or upwards in a gravitational field, and degree and nature of expression of mutant genes.

Two extensive experiments failed to produce any genetic change. The first concerned the expression of the mutant ocelliless in *D. subobscura* (Maynard Smith and Sondhi 1960). Mutant flies lacked one or more of the ocelli (simple eyes) and bristles on top of the head. Selection could alter which structures were present and which absent: for example, a population was produced whose members usually had the two posterior ocelli but not the front one. However, an experiment in which all selected parents had the left ocellus but not the right one produced no change from the initial state in which the probabilities of the presence of the left and right ocellus were equal (although selection did increase the frequency of asymmetric flies but left- and right-handed ones equally).

A second trait for which there seems to be little genetic variance in *Drosophila* is the sex ratio. In a particularly large-scale experiment, Toro and Charlesworth (1982) selected for a male-biased sex ratio, by counting many sibships in each generation, and continuing from those with the highest proportion of males. The proportion of males in the initial population was 0.5004, and after nine generations it was 0.4993.

These two experiments show that there are traits that vary, but for which there is no heritable variation. They are, however, exceptional.

mutation? This is usually obvious from the rate of response, which is greatest in the first few generations. More direct evidence is that we do not get an equivalent response if we start from a genetically homogeneous population (produced, for example, by prolonged inbreeding see p. 100).

A second source of information about the genetic variability of natural populations comes from the study of protein variability, particularly by electrophoresis. A piece of tissue (or, in the case of small organisms like *Drosophila*, the whole animal) is ground up to disrupt the cells, and centrifuged to remove insoluble material. The soluble proteins are then placed in a well at one edge of a gel immersed in an electric field. The proteins travel through the gel at a rate depending on their charge and configuration. Those present in large amounts can then be made visible by using a protein stain, but enzymes are present in too low a concentration for this to be possible. However, a specific enzyme can be made visible by supplying an appropriate substrate and a dye that will indicate the occurrence of the relevant reaction. Suppose, then, that a population is segregating

for a pair of alleles at a locus determining a monomeric enzyme. If the resulting enzymes differ sufficiently to be separable on a gel, homozygous individuals will be recognizable by the presence of single bands, in different positions for the two types. Heterozygotes will show both bands. If the enzyme is dimeric, the heterozygote will show the two parental bands, plus a third intermediate band representing the hybrid dimer. These interpretations are often clear-cut, but need checking by breeding experiments in doubtful cases.

Not all genetic variability at a locus will be revealed in this way. Usually, only changes that alter the charge on an amino acid will be detected: the actual proportion will depend on the skill and effort expended. In typical studies perhaps one quarter of all variants are detected, but this figure should not be taken as more than a rough guide. Given a body of data on a number of loci, two measures of population variability can be made:

1. Proportion of loci 'polymorphic' in the population. A locus is treated as polymorphic if the frequency of the commonest allele is less than 0.99. The value chosen is arbitrary, but some choice has to be made. Thus, if one looked for long enough, one would find rare alleles at every locus. The choice of 0.99 is intended to ensure that deleterious alleles maintained by recurrent mutation are not included.
2. Average proportion of loci heterozygous in an individual. This proportion is necessarily lower than the proportion polymorphic. For example, suppose that at some locus there are two selectively neutral alleles, with frequencies 0.2 and 0.8. This locus is polymorphic, but, if mating is random, only $2 \times 0.2 \times 0.8$, or 32 per cent of individuals are heterozygous.

Some data on electrophoretic variability in natural populations are given in Box 4.2.

Ultimately, the only way of finding out what fraction of the total genetic variability in a population is detected by a technique such as electrophoresis is to determine the DNA sequence of a sample of genes at a locus from that population. This is a less laborious task than it used to be. A classic study is Kreitman's (1983) sequencing of 11 allelic genes from *Drosophila melanogaster* coding for the enzyme alcohol dehydrogenase. There is an almost universal polymorphism for two electrophoretically separable alleles, *F* and *S*, at this locus, caused by the substitution of threonine by lysine at a particular site. Kreitman sequenced five *Adh-F* and six *Adh-S* alleles. He found no variation among his 11 genes (other than the *F-S* polymorphism itself) that would cause a change in the protein coded for. This suggests that the variability detected by electrophoresis is a substantial part of the total protein variability.

However, he did discover a number of previously hidden polymorphisms, including 13 base substitutions in the coding region of the gene that, because of the redundancy of the code, do not alter an amino acid, and 29 further substitutions in

Box 4.2— Electrophoretic Variability

Table 4.1 summarizes a large body of data on the electrophoretic variability of natural populations. The main conclusion is that most species show extensive variability: usually more than 10 per cent of the loci examined were found to be polymorphic. The main trend observable from the table is that vertebrates tend to be less variable than invertebrates. A more detailed look at the data (Fig. 4.1) shows that the variability of most invertebrates is rather similar to that of vertebrates (average heterozygosity less than 0.1), but that a substantial fraction show very high variabilities.

Table 4.1

Electrophoretically detectable variability in animals and plants (from Futuyma 1986, data mainly from Selander 1976).

	Number of species examined	Average number of loci per species	Average proportion of loci Polymorphic per population	Heterozygous per individual
Insects				
<i>Drosophila</i>	28	24	0.529	0.150
Others	4	18	0.531	0.151
Haplo-diploid wasps	6	15	0.243	0.062
Marine invertebrates	9	26	0.587	0.147
Marine snails	5	17	0.175	0.083
Land snails	5	18	0.437	0.150
Fish	14	21	0.306	0.078
Amphibians	11	22	0.336	0.082
Reptiles	9	21	0.231	0.047
Birds	4	19	0.145	0.042
Rodents	26	26	0.202	0.054
Large mammals	4	40	0.233	0.037
Plants	8	8	0.464	0.170

People have searched for other factors associated with electrophoretic variability. Only three such factors are well established:

1. Monomeric enzymes are more variable than multimeric ones.
2. Proteins of high molecular weight are more variable than smaller proteins.
3. Large populations are more variable than small ones. For example, populations that have been confined for a long time to a cave or island tend to be uniform. A striking case is the cheetah. All individuals examined from

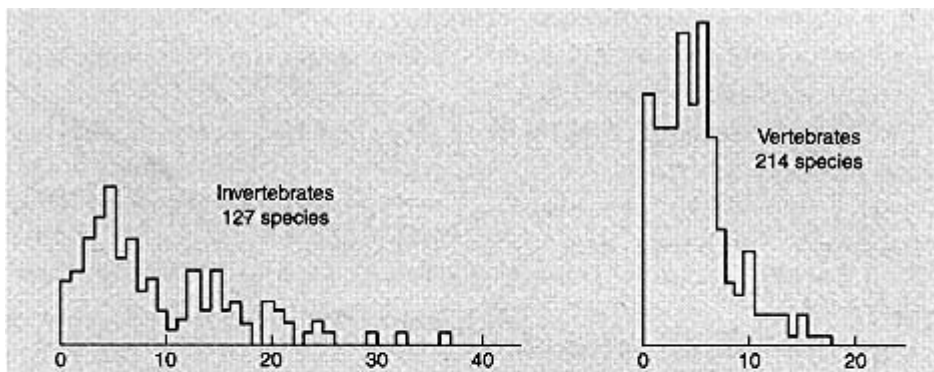


Figure 4.1 Distribution of average heterozygosity for species of invertebrates and vertebrates. Only species in which 20 or more loci were examined are included. (From Nei and Graur 1984.)

East Africa were monomorphic at all 52 loci studied (O'Brien *et al.* 1985). Skin grafts between individuals were accepted, as they are between members of an inbred line of mice. This degree of uniformity is greater than would be predicted from the present numbers of cheetahs (although that number is not large), and suggests a small bottleneck of numbers in the recent past.

These three generalizations are consistent with the 'neutral mutation theory', discussed in Chapter 8. For further details, see Nei and Graur (1984).

non-translated regions of the chromosome ('introns' and flanking regions see p. 203).

A technique of detecting genetic variability that is increasingly used in population genetics, because it is more sensitive than protein electrophoresis, but less time-consuming than DNA sequencing, is the use of restriction endonucleases. There are enzymes that cut DNA at particular sites. The method is discussed further on p. 87.

Data from electrophoresis and from artificial selection tell the same story: natural populations of sexual species are genetically variable. The rest of this chapter is concerned with the processes that cause and maintain that variability. I first discuss mutation, the origin of all genetic variation; I then consider processes that can maintain two or more alleles at intermediate frequencies in a population.

Mutation

The Nature of Mutation

A mutation can be defined as any change in the base sequence of the DNA in the genome. Mutations may involve:

1. Base substitution: the replacement of one base by another.
2. The insertion or deletion of single bases. Such mutations involve a 'frame shift' in the process of translation.
3. Inversion of a section of DNA. Figure 4.2 shows that this results in the insertion into the transcribed strand of an inverted portion of the complementary strand, and is therefore likely to be lethal if it occurs in an non-essential region of the DNA. Viable inversions presumably involve breaks in non-essential regions, and are such that the instructions concerning which strand is to be transcribed are preserved.
4. Duplication or deletion of a section of the DNA.

These kinds of mutation occur at different rates, and are differently affected by mutagenic agents, but there is no reason to think that there is any constraint *at the level of the DNA* on what mutations can occur. Some kinds (for example, Type 2) are almost certain to be lethal if they occur in an essential region of the DNA. But this is a constraint on what mutants will survive, and not on what will occur in the first place. In any particular species there will be severe limitations on the kinds of adult mutant phenotypes one observes: one will not find adult insects, or vertebrates, with an occluded gut. But these constraints arise from physiology and development, and not from the mutational process itself: so far as we know, there is no class of DNA sequence that cannot arise by mutation.

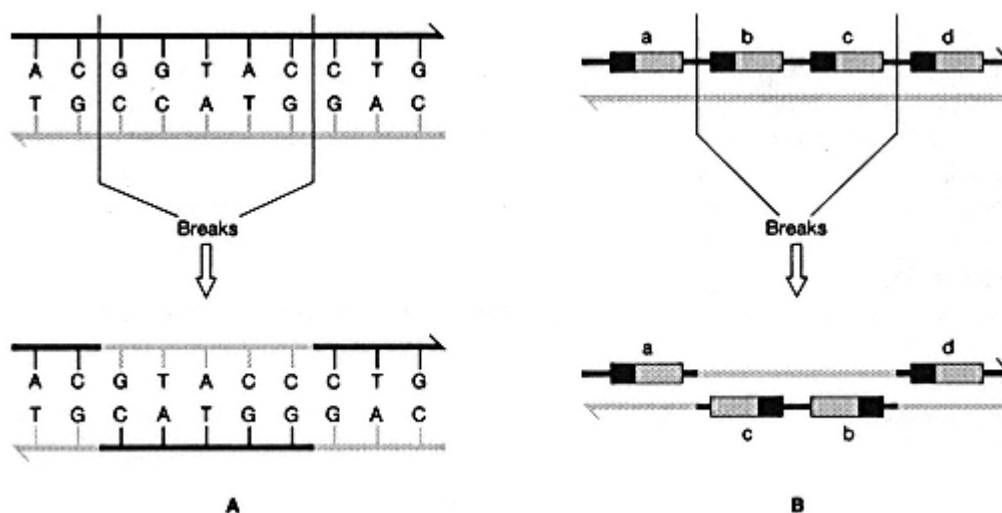


Figure 4.2

Intragenic (**A**) and chromosomal (**B**) inversions. In both diagrams, the transcribed DNA strand is indicated by a bold line. In **A**, the breaks occur within a transcribed region, and the resultant protein is almost certain to be non-functional. In **B**, the breaks occur in non-transcribed regions. Each of the four transcribed regions, a, b, c, and d, has its own signal (indicated by a black region) indicating the start of transcription. The inversion, therefore, need have no phenotypic effect.

The Balance Between Mutation and Selection

It is common sense that most mutations that alter fitness at all will lower it. Random changes are as likely to lower the fitness of organisms as they are to reduce the efficiency of automobiles or computer programs. I start the study of mutation, therefore, by considering the distribution of deleterious mutations in populations, first in a theoretical model, and then empirically.

How many deleterious mutations do we expect to find in natural populations? We answer this question by noting that, in the long run, the rate at which new deleterious mutations occur must equal that at which mutations are eliminated by selection. Consider a large random-mating population of size N . At some locus, a symbolizes the wild-type allele and A a mutant allele (or class of alleles). If p is the frequency, in new zygotes, of A , we have:

Genotype	aa	Aa	AA
Fitness	1	$1 - hs$	$1 - s$
Number of zygotes	Nq^2	$2Npq$	Np^2

First we calculate the number of A genes, present in these zygotes, that are lost by selection. There are $2Npq$ Aa zygotes, of which a proportion hs die selectively. Each death eliminates one A gene. Hence $2Npqhs$ A genes are eliminated in heterozygotes. There are Np^2 AA homozygotes, of which a proportion s die; each death eliminates two A genes. In total, then

$$A \text{ genes lost} = 2Npqhs + 2Np^2s = 2Nps(qh + p).$$

Also, in each generation, new A genes arise by mutation. Let the mutation rate from a to A (i.e. the probability, per generation, that an a gene will mutate to A) be u . There are $2Nq$ a genes in the population, and hence $2Nqu$ new A genes arise by mutation in each generation. (We ignore back mutation from A to a for two reasons. First, there are relatively few A genes. Secondly, there are many possible mutation changes that will convert a functional (a) gene into a non-functional one (A), but there may be only one that will restore function to A .)

At equilibrium the number of new mutations equals the number eliminated. That is,

$$\begin{aligned} 2Nqu &= 2Nps(qh + p), \\ \text{or } qu &= ps(qh + p). \end{aligned} \quad (4.1)$$

If A is fully recessive ($h = 0$), $qu = p^2s$, or, since $q \cong 1$,

$$p \cong \sqrt{(u/s)}. \quad (4.2)$$

If A is partially or wholly dominant, $p \cong qh$, so

$$p \cong u/sh. \quad (4.3)$$

In the special case of a dominant lethal, $sh = 1$, and hence $p = u$. All mutants in the population in this generation arose in the last one.

What fraction of the population dies each generation because of deleterious mutation at a locus? This fraction is known as the **mutational load**.

Consider first a fully recessive gene. From Equation 4.2, the fraction of the population affected is $p^2 = u/s$. Of these, a proportion s die. Hence the mutational load is $u/s \times s = u$. For a partially or wholly dominant deleterious gene, homozygotes are very rare, and we need only take account of the heterozygotes, whose frequency is approximately $2p = 2u/sh$. Of these, a proportion sh die. Hence the mutational load is $2u$.

Note that the mutational load at a locus depends only on the mutation rate, and not on the selective coefficient. This conclusion could have been reached without going through the algebra of the preceding paragraphs. The essential point is that, at equilibrium between mutation and selection, the number of new mutations occurring in the population in a generation equals the number eliminated by selection. The number of new mutations in a diploid population of size N is $2Nu$ per generation. If the mutation is dominant, each selective death removes only one mutant gene, so $2Nu$ deaths must occur every generation. For a recessive, each death removes two genes, so Nu deaths must occur. Note that this argument, originally due to J.B.S. Haldane, has not assumed random mating. The mutational load is not only independent of s ; it is also independent of the mating system.

Can we combine the mutational loads from many loci, to obtain a total load? Suppose, for example, that the load at a single locus is 0.001, and that there are 1000 loci. Does this imply a total load of $1000 \times 0.001 = 1$; that is, does mutation cause the death of the whole population? Things are not quite as bad as that. Thus the chance that an individual will not die because of mutation at a particular locus is 0.999. If we can assume that his chance of dying because of mutation at one locus is independent of his chances of dying because of mutation at other loci, then his chance of surviving is $(0.999)^{1000} = 0.368$, so that the total mutational load is 0.632. But can we assume that loci act independently in this way? Only if the following two things are true:

1. The presence of a mutation at one locus is independent of the presence of mutations at others: this is usually a reasonable assumption.
2. The effect of mutations on fitness are multiplicative: that is, the fitness of an individual with several mutations is the product of the fitnesses of individuals carrying each mutation on its own. This assumption is reasonable for lethal or serious mutations, but doubtful for less serious ones. Thus suppose that individuals with only one or two mutations suffer no serious loss of fitness, but that the presence of a larger number does seriously reduce fitness: that is, they act **synergistically**. Then the total mutational load will be lower than that based on the multiplicative assumption, because each selective death removes a

larger number of mutations. There is some evidence that mildly deleterious mutations do act synergistically.

The idea of a genetic load can be related to that of mean fitness, $\bar{W} = \sum P_i W_i$, where P_i is the frequency of the i th genotype, and W_i is its fitness. Thus let the load be L . Then the fraction of the population surviving, relative to the fraction that would survive if all had the fittest genotype, is $1 - L$. This can be written

$$\begin{aligned} 1 - L &= \frac{\bar{W}}{W_{\text{MAX}}}, \\ \text{or} \quad L &= \frac{W_{\text{MAX}} - \bar{W}}{W_{\text{MAX}}}. \end{aligned} \quad (4.4)$$

In this form, the idea of a genetic load can be applied in contexts other than recurrent mutation (see, for example, Problems 6, 7, and 8).

Deleterious Mutations in Natural Populations.

Much of what we know about deleterious mutations in natural populations depends on *Drosophila*, and in particular on the technique of extracting a chromosome from an individual, and producing a population of flies homozygous for replicas of that chromosome. This technique, invented by H.J. Muller, is shown in Fig. 4.3. It depends on:

1. the fact that there is no recombination in male flies; and
2. the possibility of eliminating (or at least reducing to a very low level) recombination in the female by introducing complex inversions.

In this way, one can ensure that a single chromosome is replicated, if necessary for many generations, without possibility of recombining.

In the final generation of Fig. 4.3, if there were no viability differences we would expect approximately $1 + 2M$ phenotypes. If there is a lethal mutation on the 'wild-type' chromosome, we expect no wild-type progeny. The experiment therefore provides a way of measuring what fraction of chromosomes, extracted from a wild population, carry lethal mutations. Measurements of this kind have been made on a number of *Drosophila* species. When allowance is made for the fact that, by chance, some 'lethal' chromosomes carry more than one lethal gene, and for the fact that the measured chromosome is only part of the whole genome, it turns out that an average of one lethal (or almost lethal) mutation per genome is typical (the estimates ranged from 0.33 to 2.37).

It is critical that, in these experiments, what we demonstrate is the lethality of a chromosome when homozygous: we can only deduce that the lethality is caused by genes at one or a few loci. There is no way we can produce flies that are homozygous at one gene locus, but segregate randomly elsewhere.

Given that estimates of the percentage of lethal second chromosomes in

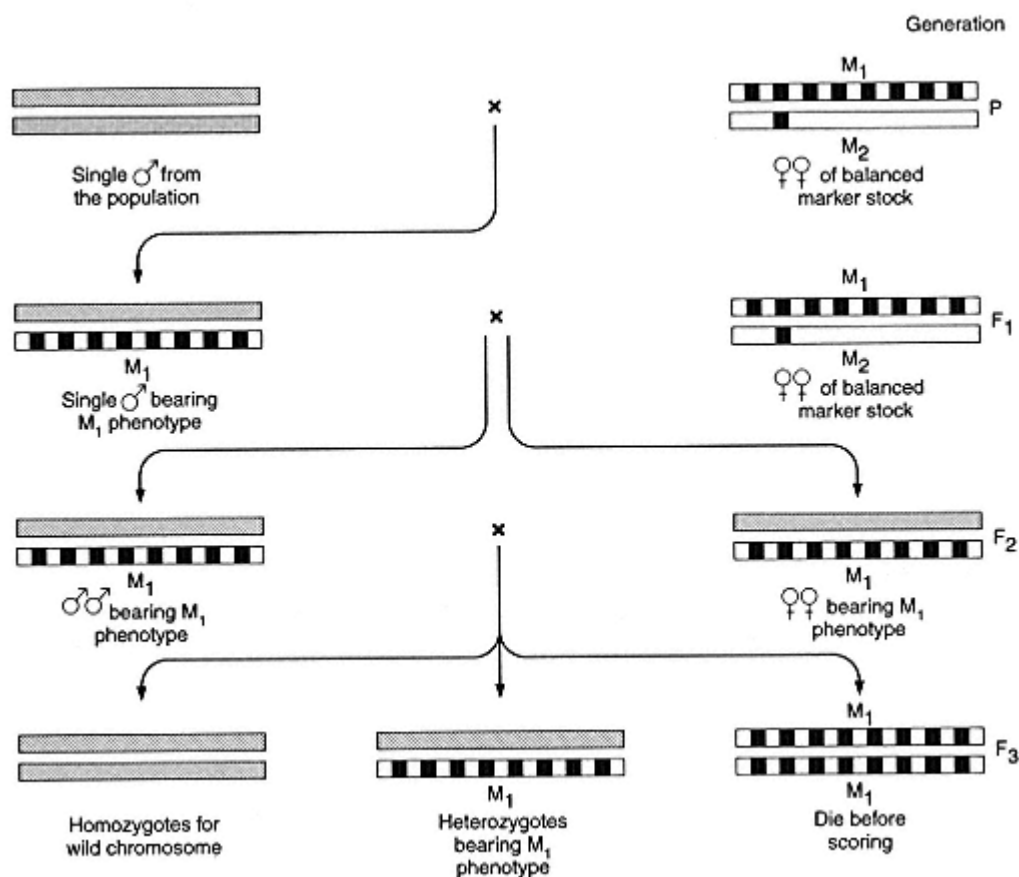


Figure 4.3

Replication scheme for sampling a chromosome from a population and providing a large number of individuals homozygous for that chromosome (from Lewontin 1974).

populations of *Drosophila melanogaster* varied from 12.3 to 61.3, we can ask how many gene loci there are on that chromosome at which lethal mutations are possible: that is, how many 'essential' genes are there? There are two quite different ways of answering this question:

1. We can look at the 'allelism' of the lethals we find. If the heterozygote between two lethal chromosomes is itself lethal, then the two chromosomes carry allelic lethals. To use this method, clearly we must have lethals extracted from different populations. You will understand this method better when you have solved Problem 2.
2. We can induce lethals in a particular chromosome region, and find out how closely they are packed. For example, Hochman (1971) estimated that there are 36 loci susceptible to lethal mutation on the fourth chromosome of *D. melanogaster*, which is about 2.6 per cent of the euchromatic length of the genome.

Fortunately, these two methods give approximately the same answer: there are 1000 essential gene loci in *Drosophila*. Since there is about one lethal mutation per genome, this implies a frequency, per lethal, of 0.001.

What of the viabilities of flies homozygous for chromosomes that do not carry lethal genes? To answer this question, it is not enough to count offspring in a cross of the kind shown in Fig. 4.3, and compare the results with the expected 1:2 ratio, because $+ / M$ flies are also of reduced viability. We therefore proceed as follows. Let $+^1$ and $+^2$ be two wild-type chromosomes. We compare the offspring ratios in crosses of the kind $+^1 / M \times +^1 / M$ (giving homozygous viabilities) with ratios in crosses $+^1 / M \times +^2 / M$ (giving heterozygous viabilities). The method is illustrated in Problem 9. Figure 4.4 shows the results of such an experiment. It is clear that, not only are there lethal chromosomes, but also that most non-lethal chromosomes have lowered viability.

Again, there is no way of telling whether the reduced viability of a given chromosome is caused by a few mutations, or by many mutations of small effect. Also, it is not clear from the figure whether the overlap between the two distributions represents experimental error (only 200 flies were counted from each cross), or whether there are chromosomes of high viability in the homozygote. Population cage experiments help to answer this question. If a cage is set up containing $+^1 / M$ flies only, then the lethal mutant M will be eliminated if $+^1 / +^1$

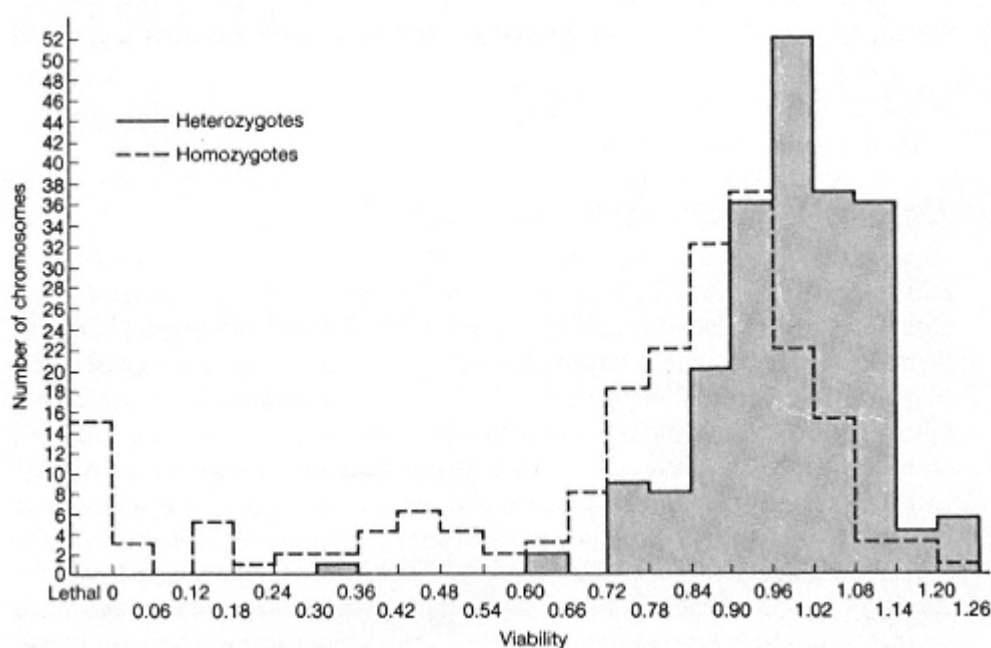


Figure 4.4
Relative viabilities of second chromosome homozygotes, and heterozygotes, in *Drosophila pseudoobscura* (data from Dobzhansky *et al.* 1963).

is fitter than $+^1 / M$. In fact, such cages continue to segregate for M . However, if the cage is set up with many wild-type chromosomes, $+^1 / M$, $+^2 / M$, $+^3 / M$, and so on, the mutant M is rapidly eliminated. But it has so far proved impossible to find a wild-type chromosome that, in homozygous condition, is fitter than $+^1 / M$. Notice that in population cage experiments not only viability but other fitness components (e.g. fecundity, mating success, longevity) are relevant. Probably for this reason the average fitness of $+^1 / +^1$ homozygotes, compared to $+^1 / +^2$ heterozygotes, is about 0.25, which is much lower than the viability data of Fig. 4.4 suggest.

To conclude, both lethal and deleterious mutations are common in natural populations. Very approximately, *Drosophila* populations carry one lethal mutation per genome, and there are 1000 essential loci, so that a typical frequency for a lethal recessive is 0.001. Most chromosomes from natural populations are of low fitness in homozygous condition, and often of very low fitness.

The Rate of Mutation.

The last section was concerned with the 'standing crop' of deleterious mutations. I now turn to the problem of estimating the rate at which new mutation is occurring. There are in fact at least four different kinds of rate that one might estimate:

- (1) the rate of base substitution, per base, per replication;
- (2) the rate at which new mutations occur at a gene locus, per generation;
- (3) the rate at which lethal or deleterious mutations accumulate on a chromosome;
- (4) the rate at which new phenotypic variance is generated by mutation.

These will be considered in turn.

The Rate of Base Substitution

Suppose that, in a bacterium such as *E. coli* that is able to live in the absence of a particular nutrient say tryptophan, one isolates a mutant unable to grow in the absence of tryptophan, because it lacks a particular enzyme, tryptophan synthetase; this is referred to as **forward mutation**. It is then possible to measure the frequency in a bacterial population of cells carrying **back mutation**, by plating out, say, 10^9 cells and counting the relatively small number of growing colonies, each derived from a single cell carrying a back mutation. One must then convert such a measure of the frequency of cells carrying a back mutation into an estimate of the rate of back mutation per cell division: one simple way of doing this is illustrated in Problem 10. It is also necessary to check that a growing colony does consist of cells in which the original forward mutation in the gene for tryptophan synthetase has been reversed, and not cells in which the defect has been overcome in some other way.

If the original forward mutation is known to be a change in a single base, say

A → T, then a measurement of back mutation estimates the rate of mutation T → A per cell division (that is, per replication). One conclusion from such experiments is that rates are rather variable, and depend on the neighbouring bases. However, most values lie in the range 10^{-8} - 10^{-10} . The rate is similar for prokaryotes and simple eukaryotes. Data on higher eukaryotes (e.g. mammals) are harder to obtain, but suggest a similar value.

It is worth asking how such an astonishing degree of accuracy is achieved. There are three stages. In the first, a polymerase enzyme inserts a new base, with a probability of error of about 1 in 10^4 . This is the only stage in RNA replication, and hence this is the per base mutation rate. In DNA replication, there is then a **proof-reading** stage: an enzyme bound to the polymerase recognizes wrong bases, and removes them, giving the polymerase a second chance to insert the correct base. The chance of an error escaping detection at this stage lies between 1/100 and 1/1000. Finally, there is a third stage, **mismatch repair**. Immediately after replication, when the new and old strands are still distinguishable (for example, because the new strand is not yet methylated), an enzyme recognizes any remaining mismatch. If one is detected, the new strand is removed, for some hundreds of bases, and resynthesized. The chance of an error escaping detection at this stage is about 1/100. Combining these three stages gives an estimated mutation rate of between 10^{-8} and 10^{-9} . Actual measurements suggest that accuracy is better than this by a factor of about 10; this is not a serious disagreement for such a difficult measurement.

The Rate of Mutation at a Locus

By a sufficient expenditure of hard work, we can record the frequency with which mutations occur at specific loci. For example, Schlegel and Dickie (1971) found 25 new coat-colour mutations at five loci in the mouse, in over two million gametes tested, giving an average rate of visible mutants of 1.1×10^{-6} per locus, per generation. Two separate estimates of the rate at which new electrophoretically detectable mutations occurred in *Drosophila* were close to 4×10^{-6} , per locus, per generation.

These estimates cannot be very accurate, but they are clearly much higher than those for base substitution rates. There are two reasons for this. First, a 'generation' in a mouse or a fly represents 20-30 cell divisions (remember, $2^{20} \approx 10^6$, so 20 cell divisions can give a million cells). Thus the values just quoted for mice and *Drosophila* suggest a rate per cell division of between 10^{-6} and 10^{-7} . This agrees rather well with estimates of the rate per cell division in bacteria of mutations destroying the function of genes specifying enzymes. Hence, it seems that mutation rates per cell division are similar in prokaryotes and eukaryotes.

A second reason why these per-locus rates are much higher than base substitution rates is that many different base substitutions would be recorded as

mutations. Thus suppose an enzyme contains 200 amino acids. The corresponding gene contains 600 coding bases, each of which can mutate in three ways, giving a total of 1800 different possible base substitutions. Not all of these would be recorded as mutations. Because of the redundancy of the code, some would be 'synonymous', and would make no alteration in the protein produced. Some that did alter the amino acid composition of the protein would not alter its function in a recognizable way. But probably between 100 and 1000 of the possible substitutions would be detectable, so that measured rates per gene locus, per replication, should be greater by two or three orders of magnitude, as in fact they are.

Before leaving the problem of per locus mutation rates, it is worth asking whether we cannot use an indirect method of estimation, based on Equation 4.1. Thus data suggest a frequency in natural populations of *Drosophila* of recessive lethals per locus of the order 0.001. If $u = p^2$, this implies a per generation mutation rate of 10^{-5} . Since this agrees reasonably well with the direct estimates, we can rest content, but there are good reasons for distrusting such indirect estimates, at least for autosomal recessive mutations. First, we do not know that the mutations are fully recessive. As is clear from Equation 4.1, if the mutant is only slightly deleterious (or beneficial) in heterozygotes, this would substantially alter our estimate of u . Secondly, the equation assumes random mating. If mating between relatives is common, this also would alter our estimate of u . For these reasons, it is best to regard any agreement between indirect and direct estimates as a lucky accident.

These objections do not apply to dominant lethal mutations, and are less serious for sex-linked mutations. Problem 3 gives a numerical example for chondroplasia in humans. The estimated mutation rate is 4.3×10^{-5} . Although there is no reason to doubt this estimate, there cannot be many dominant mutations with rates as high as this.

The Accumulation of Deleterious Mutations

By a simple extension of the breeding system of Fig. 4.3, one can maintain a single chromosome heterozygous and free of crossing over for many generations, and then measure its viability in homozygous condition. Mukai (1969) maintained 101 second chromosomes of *D. melanogaster* for 60 generations in this way, testing them for viability at intervals. As expected, some chromosomes accumulated lethals or sub-lethals: these are omitted from the analysis. (In fact, Mukai estimated a rate of accumulation of lethals of 0.0063 per generation: if we assume 300 loci susceptible to lethal mutation, this implies a per locus rate of 2.1×10^{-5} , which agrees well with the per locus rates quoted in the last section.) The results are plotted in Fig. 4.5. After 60 generations, the mean homozygous viability of chromosomes lacking well-defined lethals had fallen to about 50 per cent of its initial value.

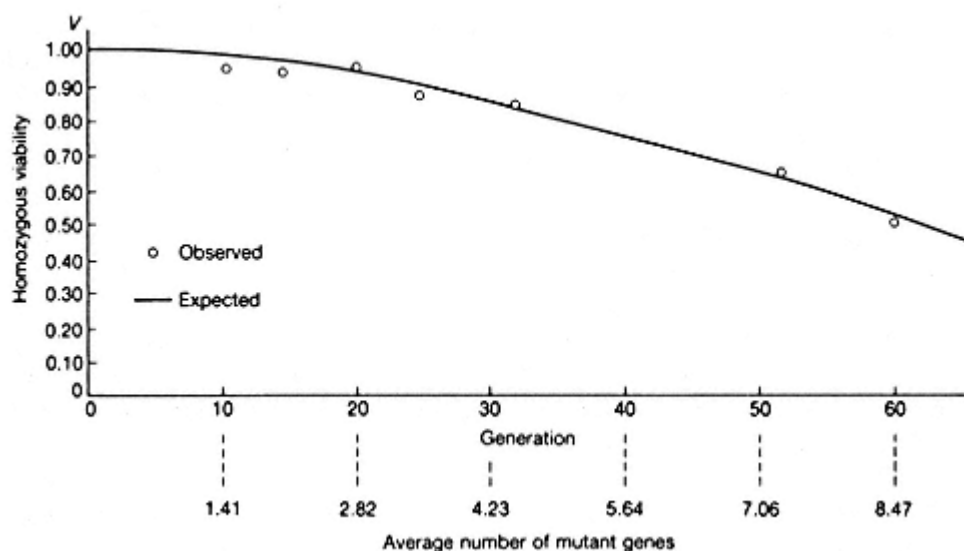


Figure 4.5

Average viability of homozygotes for *D. melanogaster* chromosomes, kept in heterozygous condition for varying numbers of generations. Lethal and sub-lethal chromosomes are omitted.

The average number of mutant genes per chromosome was estimated from the variance in viability between lines. The full line gives the expected viability, assuming a mutation rate of 0.14 per chromosome, per generation. (After Mukai 1969.)

Can we decide whether this decline was caused by mutations at many loci on each chromosome, each of small effect, or to relatively few mutations, each of large effect? The only clue we have is the variability between chromosomes. Thus if large numbers of mutations at many loci are responsible, we would expect all lines to respond in much the same way, whereas if few mutations are involved, some chromosomes will be free of mutation and others will have several, and there will be appreciable variation in viability. Hence, by using the observed variability between lines, and assuming that all mutations have equal effects on viability, one can estimate the most plausible number. This best estimate indicates a mutation rate of 0.14 per generation, and an average number of eight mutations per chromosome after 60 generations. If this is correct and the argument is reasonable it implies that the rate of slightly deleterious mutations is over 20 times higher than the rate of lethal mutations.

The Generation of Phenotypic Variation by Mutation

If one extends the breeding system of Fig. 4.3 to include all the main chromosomes simultaneously, it is possible to construct an 'isogenic line', homozygous at almost all loci. Such lines are usually sick and hard to maintain. If all differences between flies were caused by genes, then the members of an isogenic line would be pheno-

typically identical. Of course they are not: indeed, they are often more variable in morphology than wild populations. However, they do conform to theoretical prediction in one important respect—they lack *heritable* variation: for example, if one selects and mates together the largest flies, their offspring are on average no larger than the rest of the population. How rapidly does mutation restore heritable variation to an isogenic line? Clayton and Robertson (1955) estimated that between 0.1 per cent and 0.2 per cent of the variation typical of a natural population is generated by mutation in each generation. In other words, it would take between 500 and 1000 generations for mutation to restore the original genetic variability for phenotypic traits such as size or bristle number. As will be explained in Chapter 6, this observation is somewhat paradoxical, since it seems to require a higher per-locus mutation rate than is suggested by other types of measurement.

We can now summarize our conclusions about mutation rates, although it is important to remember that these are based on rather few experiments, and still fewer organisms. In particular, the reliance on the chromosome-manipulating techniques available in a single species *Drosophila melanogaster*, is worrying. With these reservations:

1. The error rate per base per replication is of the order of 10^{-10} . There is no reason to think it is very different in prokaryotes and eukaryotes.
2. The rate of mutation producing visible or lethal effects, per gene locus per generation, in higher eukaryotes is of the order of 10^{-5} .
3. Slightly deleterious mutations are more frequent than lethals, and perhaps 20 times as frequent. A likely explanation for part of this difference is that there are non-essential gene loci at which mutations are mildly deleterious but not lethal.
4. Phenotypic variation in morphological traits is generated by mutation at a rate per generation of about 0.1 per cent of the variation typical in natural populations.

The Maintenance of Variation

We are not in a position to offer a classification of the causes of variation in natural populations:

1. *Variation selectively neutral*. Different types do not differ appreciably in fitness, and hence none has been eliminated by selection. To analyse this possibility requires that we consider finite populations, so discussion is postponed to Chapter 8.
2. *Selection-mutation balance*. Less fit types are maintained in the population by repeated mutation. For mutations at specific loci, this was discussed above on p. 55. The possibility that variation for quantitative traits is maintained in this way is discussed in Chapter 6.

3. Balance of selective forces.

- (a) Heterosis: the heterozygote at a locus is fitter than either homozygote;
- (b) frequency-dependent selection;
- (c) fitnesses vary in space and time;
- (d) selection acts differently at different levels.

Before discussing the various processes under 'Balance of selective forces' in detail, it is worth making a general point about method. If we want to show that two alleles, A and a , are maintained by selection in a population, it is usually sufficient to show that each of them will increase in frequency when rare. It is easier to show this than it is to find the frequency p , at which there is a stable equilibrium. If we can show that each allele increases when rare, we have demonstrated the existence of **protected polymorphism**.

Heterosis

Suppose that the fitnesses of the three genotypes at a locus are:

Genotype	AA	Aa	aa
Fitness	$1 - s$	1	$1 - t$

It is at once obvious that there is a protected polymorphism. Thus when allele A is rare, the population consists primarily of aa individuals, with a few Aa . Allele A then occurs in individuals of higher-than-average fitness, and therefore increases in frequency. By an identical argument, allele a increases when rare.

What is the frequency, $\hat{p}$, at which there is an equilibrium? The standard, and clumsy, way of answering this question is as follows. Define the frequency of A in generation n as p_n . Calculate (exactly as in Equation 3.5) the value of p_{n+1} in terms of p_n , s and t . At equilibrium, p does not change, so $p_{n+1} = p_n = \hat{p}$. This equation can be solved for $\hat{p}$. The only objection to the method is that the algebra is rather messy.

An alternative method is to calculate the 'fitnesses' of the genes: that is, the number of copies in the next generation of a gene in this. Thus the fitness of allele A is given by:

$$(\text{probability that } A \text{ finds itself in an } AA \text{ genotype}) \times (\text{fitness of } AA) + \\ (\text{probability that } A \text{ finds itself in an } Aa \text{ genotype}) \times (\text{fitness of } Aa).$$

$$\begin{aligned} \text{That is,} & \quad W(A) = p(1 - s) + 1 - p, \\ \text{and similarly,} & \quad W(a) = p + (1 - p)(1 - t). \\ \text{At equilibrium,} & \quad W(A) = W(a), \text{ or } p = t/(s + t). \end{aligned} \quad (4.5)$$

This possible method of maintaining variability has been familiar since the early days of population genetics. The classic example was sickle-cell anaemia. A mutant

of the β chain of haemoglobin in humans, symbolized S , has the following properties:

S/S homozygotes die of anaemia,

$S/+$ heterozygotes do not suffer from anaemia, but are resistant to malaria,

$+/+$ homozygotes are susceptible to malaria.

It follows that, in regions where malaria is a significant selective force, the heterozygote is the fittest of the three genotypes. In some malarial regions of Africa, the S allele is present in frequencies as high as 15 per cent.

Until the development of electrophoretic methods, however, it was rarely possible to recognize all three genotypes at a locus, so the prevalence of heterosis was hard to evaluate. When extensive electrophoretic data became available, it was widely expected that many other examples of heterosis would be discovered. This expectation was strengthened by the fact that it is physiologically plausible that a heterozygote, with two enzymes catalysing the same chemical reactions, but with different rate constants, specificities, or dependence on physical conditions, might be fitter than either homozygote. This expectation, however, has not been realized, and this has led most population geneticists to conclude that single-locus heterosis is a rare phenomenon. I am not convinced. Thus, it was clear from the outset that we could not expect to find many polymorphisms with the large fitness differences characteristic of sickle-cell anaemia. Fitness differences of 1 per cent or less would be quite sufficient to account for genetic variability, but would not be easily demonstrated. The matter is discussed further in Box 4.3.

Box 4.3—

Further Data on Heterosis

The best studied example of heterosis, by Dobzhansky and his colleagues, concerns variation in chromosome order in *Drosophila pseudoobscura*. All natural populations of this species are polymorphic for gene order on chromosome III. A number of gene orders, with names like STANDARD, ARROWHEAD, and CHIRICAHUA, are found. Individuals can only be classified for gene order when they are larvae, by examining the salivary gland chromosomes. There is good evidence that heterosis accounts for the variability. The strongest comes from studies in population cages, in which flies can be maintained for many months, and samples of larvae can be taken from time to time and classified for gene order. A typical experiment is shown in Fig. 4.6. A population approaches the equilibrium frequency from above and below, demonstrating that the equilibrium is stable. Adult flies taken from the cage can be classified by a breeding test: thus if a fly is mated to one of known chromosome type, and its larval offspring characterized, the genotype of the tested fly can be deduced. When this was done, an excess of heterozygotes was found over that predicted

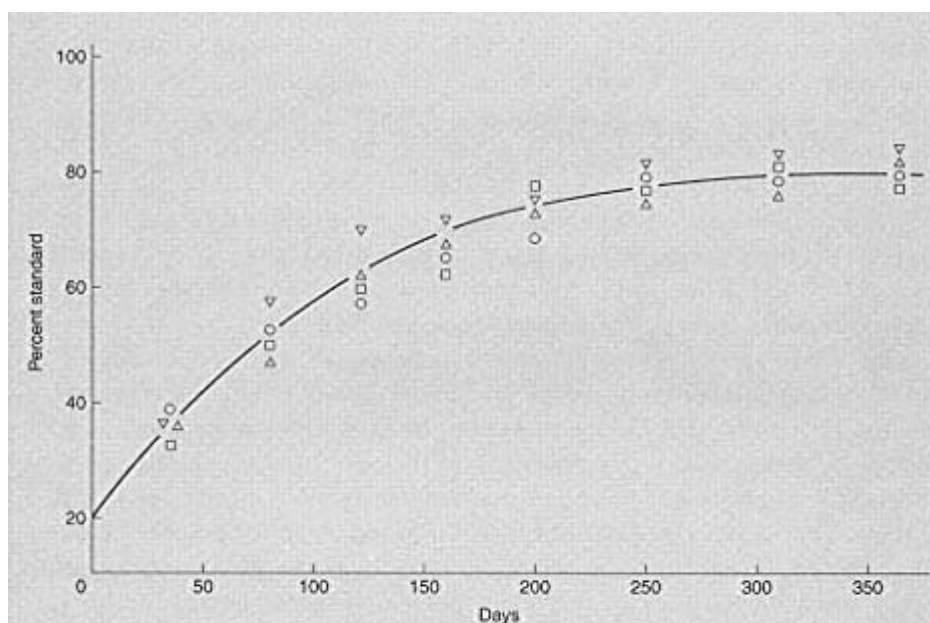


Figure 4.6 Frequency of STANDARD chromosomes in a population cage experiment with *Drosophila pseudoobscura*. Data from four replicates are shown. The populations were started with 20 per cent of STANDARD and 80 per cent of CHIRICAHUA chromosomes. (After Dobzhansky 1951.)

by the Hardy-Weinberg ratio. That this was due to selection, and not to assortative mating, was demonstrated by taking samples of eggs from the cage. If these were raised in optimal conditions, with little mortality, the resultant larvae were found to fit the Hardy-Weinberg ratio in chromosomal type.

There is no serious doubt that chromosomal polymorphism in *D. pseudoobscura* is maintained by heterosis. A few comments are needed. First, inversion polymorphism is widespread in the Diptera, but rare in most other taxa. This is because, in most taxa, meiosis in inversion heterozygotes leads to a proportion of aneuploid gametes, and hence lowered fertility. This does not happen in the Diptera, because there are no chiasmata in male meiosis, and in female meiosis any aneuploid nuclei that are formed finish up in the polar bodies, and not in the egg pronuclei.

A second comment concerns the relation between chromosomal and genetic heterosis. Almost certainly, the superior fitness of inversion heterozygotes in *Drosophila* has nothing to do with the effects of the inversion itself, but arises because different chromosome orders carry different alleles. There is electrophoretic evidence that this is so. It arises because genetic recombination between different orders is rare or absent. However, two kinds of allelic effects

on fitness could be involved (Fig. 4.7). In Type I, inversion heterozygotes are also heterozygous for heterotic alleles at various loci: in this case, inversion heterosis is caused by genic heterosis. In Type II, the advantage of inversion heterozygotes arises from interactions between loci. I have shown the simplest kind of interaction: each order carries one recessive deleterious mutation. Therefore, both gene order homozygotes have a lower fitness than the heterozygote, yet there are no individual loci that are heterotic. It is because of this second possibility that one cannot conclude from the prevalence of inversion heterosis that there is also widespread genic heterosis.

One advantage of population cage experiments is that they enable us to observe the cumulative effects of selection over many generations. One might therefore suppose that the method could be used to investigate heterosis at single loci, when selection is too weak to measure over a single generation. Such experiments would indeed be decisive if it were not for the phenomenon of linkage. The relevance of linkage is illustrated by an experiment by Jones and Yamazaki (1974) on the esterase-6 locus in *Drosophila pseudoobscura*. They found that if they established a population cage, segregating for two alleles at this locus, with only a small number of flies, there were marked changes of frequency at the enzyme locus. But if the cage was established with a large number of flies, the frequency changes were less marked. This strongly suggests that the changes observed in the first type of cage were caused by selection, not on the esterase locus itself, but on loci linked to it.

An alternative method (Mitton and Grant, 1984) of deciding whether electrophoretic variability is maintained by heterosis is to take a sample of individuals from a natural population and record for each its growth rate (or some other measure of success) and its genotype at a number of enzyme loci. A positive correlation has been observed between growth rate and number of

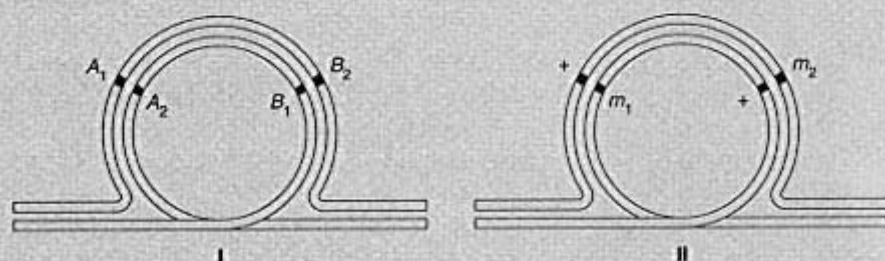


Figure 4.7 Two ways in which effects at individual loci could cause inversion heterosis. The chromosomes are shown paired, as they would be in the salivaries. In **I**, there are two heterotic loci: that is, A_1/A_2 is fitter than A_1/A_1 or A_2/A_2 , and similarly at the B locus. Of course, a single heterotic locus would be sufficient. In **II**, there are two loci, at each of which there is a deleterious recessive allele, m_1 and m_2 , respectively. This is the simplest form of interaction between loci generating inversion heterosis.

loci heterozygous in several species of oyster, in a mussel, in a salamander, in white-tailed deer, in quaking aspen, and in several conifer species. One possible explanation is that the effect is caused by the genes whose products are being recorded: that is, heterozygotes at these specific loci do indeed grow faster. A second possibility is that individuals that are more homozygous for the enzyme loci are more homozygous at other loci as well, and that it is these other loci that are relevant. This would be plausible if the populations sampled consisted of individuals that were inbred to varying degrees. This is a reasonable explanation for the tree species, in which seed dispersal is limited and pollination often occurs between neighbours, but it is implausible for oysters and mussels, which have planktonic larvae. The interpretation of these data is still controversial, but they do lend some support to the idea that single-locus heterosis is widespread.

Frequency-dependent Selection

The basic idea is very simple. If two types exist in a population, and if the fitness of each is greater when it is rare, then stable coexistence will result. This is true whether the two types reproduce asexually, or are different genotypes in a sexual population.

There are good ecological reasons for thinking that frequency-dependent selection is a major cause of genetic variability. Many of the most obvious kinds of selection are likely to be frequency-dependent in their effects. For example:

1. *Disease.* As Haldane was the first to point out, parasites will evolve so as to attack most effectively the common genotypes in the host population; therefore rare host genotypes will be at an advantage. Similarly, hosts will acquire immunity to the common types of parasite, so rare types of parasites will be at an advantage.
2. *Predation.* If predators form a 'search image' of their prey, unusual individuals are more likely to escape predation.
3. *Resource utilization.* If different genotypes have slightly different resource requirements, this may favour coexistence, for essentially the same reason that species with different requirements can coexist.
4. *Behavioural variability.* In many situations the benefits of particular behaviours are frequency-dependent, and this may lead to a 'mixed strategy' being stable: examples include foraging behaviour, fighting, and male mating behaviour. In such situations, the population may become genetically polymorphic, or individuals may be flexible in their behaviour.

Direct evidence for genetic polymorphism at individual loci being maintained by frequency-dependent selection is equivocal. Most of the experiments concern

larval viability in *Drosophila*. Evidence both for and against frequency-dependent selection on enzyme loci is extensive. However, it is hard to believe that the resource requirements of larval *Drosophila* can be so finely differentiated that they can be responsible for the maintenance of a substantial part of the observed polymorphism.

Two final points are worth making. First, there is nothing artificial about assuming that fitnesses are frequency-dependent. The artificial assumption is that relative fitnesses are constant: its justification is mathematical convenience, not truth. Secondly, the most widespread and dramatic genetic polymorphism, that of sexual dimorphism, is certainly maintained by frequency-dependent selection: as Fisher pointed out, in a sexual species, whichever is the rarer sex has most children.

Rather little work has been done on genetic models in which fitnesses are frequency-dependent. The natural way to analyse such cases is to think about evolutionary change in the phenotype directly, as described in Chapter 7.

A Variable Environment.

It seems only common sense that a population should be more diverse genetically if it lives in a variable environment. However, this conclusion depends critically on the nature of the environmental variability.

Variability in Space

The first, rather obvious, point is that the environment, if it is to maintain genetic variability, must vary in a 'coarse-grained' manner. Thus if we imagine the environment to be made up of patches of different types, these patches must be large relative to the movement of an individual organism. If this were not so, each individual would experience many patches, but all individuals would, during their lives, experience the same set of conditions. Therefore, each genotype would have a fixed fitness, and the variable environment would not maintain a variable population. Clearly, an environment that is coarse-grained to a sedentary organism may be fine-grained to a mobile one.

We are concerned, then, only with coarse-grained environments. Figure 4.8 shows two models, due to Dempster (1955) and Levene (1953), which, although apparently similar, lead to different conclusions. They are worth following in some detail, because of the light they shed on the way models in population genetics are constructed and analysed.

Both models consider two alleles, A and a , at a locus (they can readily be extended to many alleles). Both assume a single random-mating population, and two kinds of patch (this, too, can be extended). In Dempster's model, fixed proportions, c_1 and c_2 , of the zygotes settle in patches 1 and 2 respectively (note that $c_1 + c_2 = 1$). Once in a patch, each genotype has a fixed absolute fitness, or chance of surviving. Survivors from each patch join a single random-mating

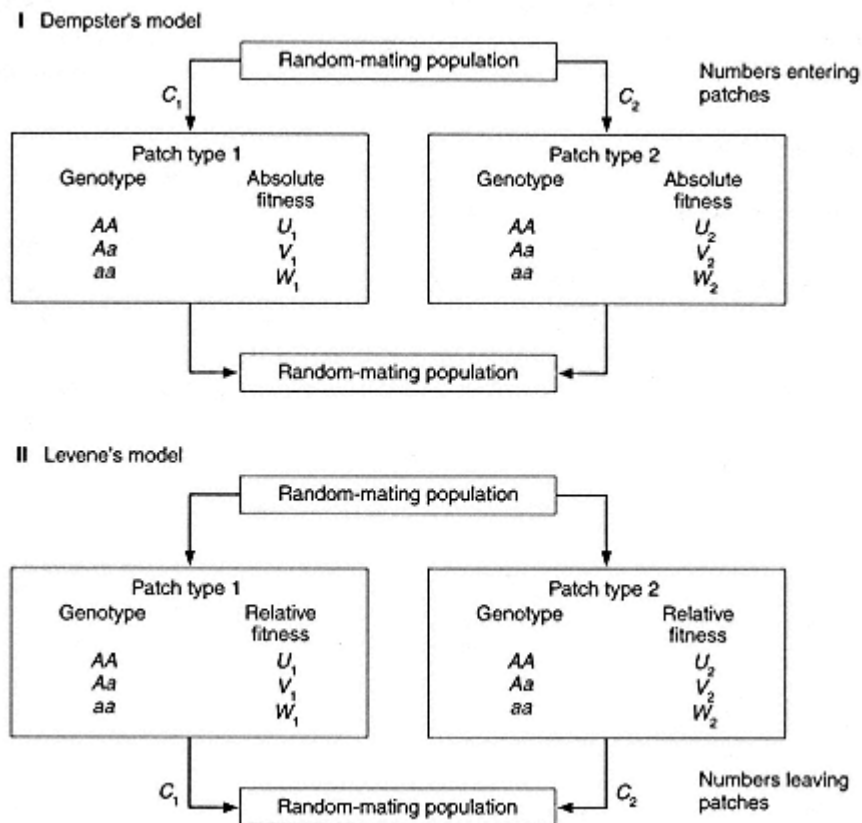


Figure 4.8
Models used to investigate whether polymorphism can be maintained in a variable environment.

population to produce, without selection, the next generation of zygotes. We can first ask, what is the fitness of genotype AA , averaged over environments. That is, if we select a random AA zygote, what is its probability, $W(AA)$, of survival? Clearly, $W(AA) = c_1 U_1 + c_2 U_2$. Similarly, $W(Aa) = c_1 V_1 + c_2 V_2$ and $W(aa) = c_1 W_1 + c_2 W_2$.

The essential feature of Dempster's model is that these genotypic fitnesses are constants, and independent of gene frequency. It shows that there can be a stable polymorphism only if the heterozygote is fitter than either homozygote. To see this, suppose that one allele A , is fully dominant in both patches: that is, $U_1 = V_1$ and $U_2 = V_2$. Then $W(AA) = W(Aa)$. Either $W(AA) > W(aa)$, in which case allele A will be fixed, or $W(AA) < W(aa)$, and a will be fixed. It will only be by pure accident, for one particular value of c_1 , that they will be equal. It is possible for Aa to be fitter than AA or aa , even if there is no heterosis in either patch. This would be so, for example, if $U_1 = V_1 > W_1$ and $U_2 < V_2 = W_2$; that is, if one homozygote is

fitter in patch 1, and the other in patch 2, and if the heterozygote is always as fit as the fitter homozygote. But it would be better to think of this as an example of heterosis arising because of the greater environmental tolerance of the heterozygote.

Dempster's model, then, suggests that environmental variability contributes little to the maintenance of genetic variability, except through the superiority of heterozygotes. In a haploid organism, there would be no polymorphism. Before leaving the model, however, it is worth asking how it supposes the population is regulated. There is no explicit assumption, but there is a tacit one that regulation acts on the random-mating population as a whole, and not on the separate patches: if this were not so, the absolute fitnesses in the patches could not be constants, as is assumed. This will become clearer when we consider the alternative, Levene's model.

In Levene's model, zygotes produced by random mating settle, in excess numbers, in the two patches. Each patch then produces fixed proportions, c_1 and c_2 , of adults to the next random-mating population: thus it is assumed that density-dependence operates separately within each patch. The fitnesses U_1 , U_2 , etc. are the *relative* probabilities of survival of the different genotypes. In Box 4.4, the model is analysed for A dominant (i.e. $V_1 = U_1$; $V_2 = U_2$); if polymorphism can be maintained in this case, it cannot be ascribed to heterosis. The method is to find the conditions for a protected polymorphism: they are

$$\begin{aligned} \text{for } A \text{ to increase when rare} \quad & \frac{c_1 U_1}{W_1} + \frac{c_2 U_2}{W_2} > 1, \\ \text{for } a \text{ to increase when rare,} \quad & \frac{c_1 W_1}{U_1} + \frac{c_2 W_2}{U_2} > 1. \end{aligned} \quad (4.8)$$

Box 4.4—

Levene's Model with Dominance

The assumptions of the model are shown in Fig. 4.8. If the frequency of allele A in the random-mating population in one generation is p , then the numbers surviving in the two patches are as shown in Table 4.2. Combining the two patches, the total number of genes is $2(c_1 + c_2) = 2$. The number of A genes is

$$\frac{c_1 U_1}{T_1} (2p^2 + 2pq) + \frac{c_2 U_2}{T_2} (2p^2 + 2pq) = 2p \left(\frac{c_1 U_1}{T_1} + \frac{c_2 U_2}{T_2} \right).$$

Hence the frequency p' of A in the next generation is

$$p' = p \left(\frac{c_1 U_1}{T_1} + \frac{c_2 U_2}{T_2} \right).$$

Table 4.2

Levene's model with dominance

	Genotype			Total
	AA	Aa	aa	
Zygotes settling in patch 1	p^2	$2pq$	q^2	1
Survivors in patch 1	$p^2 U_1$	$2pq U_1$	$q^2 W_1$	$T_1 = U_1 + q^2(W_1 - U_1)$
Numbers surviving in patch 1	$\frac{p^2 U_1 c_1}{T_1}$	$\frac{2pq U_1 c_1}{T_1}$	$\frac{q^2 W_1 c_1}{T_1}$	c_1
Numbers surviving in patch 2	$\frac{p^2 U_2 c_2}{T_2}$	$\frac{2pq U_2 c_2}{T_2}$	$\frac{q^2 W_2 c_2}{T_2}$	c_2

When p is small, $T_1 \approx W_1$ and $T_2 \approx W_2$, so

$$p' = p \left(\frac{c_1 U_1}{W_1} + \frac{c_2 U_2}{W_2} \right).$$

Hence the condition for allele A to increase when rare is

$$\frac{c_1 U_1}{W_1} + \frac{c_2 U_2}{W_2} > 1. \quad (4.6)$$

By a similar calculation

$$q' = pq \left(\frac{c_1 U_1}{T_1} + \frac{c_2 U_2}{T_2} \right) + q^2 \left(\frac{c_1 W_1}{T_1} + \frac{c_2 W_2}{T_2} \right),$$

and when q is small, $T_1 \approx U_1$ and $T_2 \approx U_2$, so

$$\begin{aligned} q' &= q(1 - q) + q^2 \left(\frac{c_1 W_1}{U_1} + \frac{c_2 W_2}{U_2} \right) \\ &= q + q^2 \left(\frac{c_1 W_1}{U_1} + \frac{c_2 W_2}{U_2} - 1 \right). \end{aligned}$$

Hence the condition for allele a to increase when rare is

$$\frac{c_1 W_1}{U_1} + \frac{c_2 W_2}{U_2} > 1. \quad (4.7)$$

If both Equations 4.6 and 4.7 are satisfied, there is a protected polymorphism.

To see what these imply, consider a numerical example, Let $c_1 = c_2 = 1/2$; that is, equal-sized patches. Also, since we are interested only in the relative fitnesses in each patch, we can take $U_1 = U_2 = 1$. Then Equation 4.8 becomes

$$1/W_1 + 1/W_2 > 2; \quad W_1 + W_2 > 2.$$

There is nothing impossible about these conditions; for example, they are satisfied by $W_1 = 2$; $W_2 = 0.5$. The conclusion differs from Dempster's because overall fitnesses are now frequency-dependent. To see this, suppose that aa is of low fitness in patch 1, and high fitness in patch 2. Then, when a is rare, aa genotypes will have little competition in patch 2, so many will survive. The point becomes still clearer if we take the extreme case, in which AA and Aa are lethal in patch 2, and aa is lethal in patch 1. It is then obvious that both alleles will be maintained.

However, unless the selective differences are large, the conditions for polymorphism are far from robust. To see this, let us take a pair of values of W_1 and W_2 and ask for what range of values of c_1 (relative frequency of patches) polymorphism can be maintained. Some values are

W_1	W_2	Range of c_1 permitting stable polymorphism
1.01	0.99	0.5-0.505
1.5	0.5	0.5-0.75

These ranges seem narrow. If conditions changed, so that the relative frequencies of the two patch types fell outside the range, then a previously stable polymorphism would become unstable.

Levene's model can be modified in several ways so as to be more favourable for the maintenance of polymorphism:

1. '*Habitat selection*'. Females contribute their offspring to the patch type in which they were raised. This could happen by a process equivalent to imprinting, or, in plants or sessile animals, because of limited dispersal.
2. '*Mating within a patch*'. In the limit, there could be two regions, each containing a separate random-mating population. If AA were fitter in one region, and aa in the other, it is obvious that both alleles would be maintained, one in each region. If so, neither population would be polymorphic: when different regions contain different genotypes, a species is said to be polytypic. There will be intermediate cases, in which some movement occurs between patches: polymorphism is then more easily maintained than if mating is random.

Variability in Time

Can polymorphism be maintained if the environment varies in time rather than space? Consider first the case of two asexual types. If the relative fitnesses vary from generation to generation, but if these fitnesses repeat themselves regularly in an n -generation cycle, it should be clear that stable coexistence is impossible. Thus if we take the fitness of Type 1 in the i th generation as W_i , relative to a fitness of 1 for Type 2, the overall 'fitness' over n generations will be $W = W_1 \cdot W_2 \cdot \dots \cdot W_n$, compared to 1 for Type 2. W will either be greater than or less than 1, and Type 1

will either be fixed or eliminated. The same conclusion holds if fitness values occur with fixed probabilities, rather than following a fixed cycle.

One might therefore be tempted to conclude that genetic polymorphism cannot be maintained in a sexual population at a locus with complete dominance, but this is not so. It can be shown that polymorphism is stable if the arithmetic mean fitness of the recessive is greater than 1, while its geometric mean fitness is less than one. This would be so, for example, if aa was 1.1 times as fit as AA or Aa in most generations, but if all recessives were killed off by an epidemic disease once every 20 generations.

Balance between Selection at Different Levels

Suppose that allele A is favoured relative to a by natural selection acting on individual organisms, but that Mendelian segregation is distorted, so that gametes from heterozygotes more often carry allele a than A . Selection is then also acting at the level of gametes, but in a different direction. A similar conflict could exist between selection on individuals, and on groups of individuals. It is shown in Box 4.5 that selection in different directions at different levels can lead to stable polymorphism, though it need not do so.

Box 4.5—

Selection at Different Levels

For simplicity, imagine that animals interact in groups of two. Let there be two kinds of individuals, A and S —these can be thought of as standing for 'altruistic' and 'selfish'. The fitness of an individual depends on its own behaviour, and that of its partner, according to the scheme of Table 4.3.

Table 4.3

Selection at different levels

	Type of pair					
	A	A	A	S	S	S
Fitness	4	4	2	5	1	1

Fitness matrix		
Partner		
A S		
A	4	2
Individual		
S	5	1

Note that, in the 'fitness matrix', the entries are the fitnesses of the individuals on the left, paired with the partner indicated above.

If pairs are formed randomly, then each type is the fitter of the two when it is rare. Thus if S is rare, it usually pairs with an A , and has a fitness of 5, whereas A has a fitness of 4. If A is rare, it usually pairs with S , and has a fitness of 2, whereas S has a fitness of only 1. Since each type increases in frequency when it is rare, there must be a stable polymorphism. (Note the similarity between this argument, and that used on p. 65 to explain the maintenance of polymorphism by heterosis.)

It is not hard to imagine circumstances in which fitnesses might vary like this. Consider, for example, watching for a predator when feeding in a flock. Suppose that a flock (of two in the example) is reasonably safe so long as at least one member is watching, but that watching wastes time that could be spent feeding, whereas a flock in which no bird is watching is seriously at risk. The relative fitnesses would then vary as shown in the table, with A representing watchers, and S birds that do not watch. Watching would be maintained in the population (even if the members of a flock were unrelated), but would not go to fixation. I return to this way of looking at natural selection and evolution in Chapter 7.

Further Reading

Lewontin, R.C. (1974). *The genetic basis of evolutionary change*. Columbia University Press, New York.

Nei, M. (1987). *Molecular evolutionary genetics*. Columbia University Press, New York. (Particularly Chapters 8 and 10.)

Problems

1. Four loci were examined electrophoretically in a population. At the first, three alleles were found, with frequencies 0.12, 0.7, and 0.18; at the second, two alleles had frequencies 0.37 and 0.63; no variation was detected at the other two loci. What is the average proportion of heterozygous loci?
2. Twenty-six recessive lethals were isolated from different populations of *Drosophila* species. None were allelic to any of the others. (a) How many combinations had to be tested to decide this? (b) Is the result consistent with the idea that there are approximately 1000 'essential' loci in *Drosophila*? (c)* Given that there are 1000 loci, what is the probability that exactly one allele pair would be found? (d) How does the existence of 'gene families', described in Chapter 11, affect estimates of the number of essential loci based on measurement of the allelism of lethals?
3. In humans, achondroplasia (a form of dwarfism with short arms and legs) is caused by a dominant gene. In one study, 10 achondroplasics were found among 94 075 births. 108 achondroplasics left 27 children, whereas 457 normal sibs left 582 children. Estimate the mutation rate at the locus.

4. The frequency of individuals homozygous for a recessive lethal gene in a randommating population is $1/10\,000$. (a) If the population is in equilibrium between mutation and selection, calculate the mutation rate. (b) Ignoring mutation, and assuming that the gene is maintained in the population by heterosis, what are the relative fitnesses of the heterozygote and the 'wild-type' homozygote?
5. Devise a breeding scheme, similar to that in Fig. 4.3, to establish a population of *Drosophila* isogenic for the X chromosome.
6. If the relative fitnesses of AA , Aa , and aa are 0.95, 1, and 0.5, what is the frequency of allele A at equilibrium? What is the genetic load associated with the locus?
7. Suppose that individuals homozygous for the sickle-cell gene S , have zero fitness, and that the frequency of S in the population is 0.1. If this is an equilibrium, what is the fitness of the normal homozygote, relative to that of the heterozygote? What is the genetic load?
8. In a diploid random-mating population, genotypes AA and Aa are dark, and aa is pale. Because of frequency-dependent predation, the relative fitness of dark individuals is $0.75 - P$, and of pale individuals is $1.5 - P$, where P is the frequency of pale individuals. What is the frequency of A at equilibrium? Is the equilibrium stable? What is the genetic load?
9. Using the breeding scheme illustrated in Fig. 4.3, the following crosses were set up, and the offspring counted: $+1/M \text{♀} + 2/M \text{♂}$ gave 467 + :201 M offspring; $+1/M \text{♀} + 1M \text{♂}$ gave 376 + :197 M offspring. Calculate the viability of $+1/+1$ homozygotes, taking the viability of $+1/+2$ as unity.
10. A mutant of *E. coli* is unable to grow in the absence of tryptophan. A single mutant bacterium was placed in each of 100 tubes, containing a medium with tryptophan. When the population had increased to 10^8 cells per tube, the cells were plated on a medium without tryptophan, and the number of growing colonies recorded. 72 tubes produced no colonies. Estimate the back mutation rate.

Computer Projects

1. Two loci, with two alleles at each (A,a and B,b) are segregating in an infinite random-mating population. The relative fitnesses at the single loci are:

AA	Aa	aa	BB	Bb	bb
1	$1 - s/2$	$1 - s$	1	$1 - t/2$	$1 - t$

The fitnesses combine multiplicatively. For example, the fitness of $aabb$ is $(1 - s)(1 - t)$. The frequencies of alleles A and B are $p(A)$ and $p(B)$. The fitness at each locus depends on the gene frequency at the other, according to the following scheme: $s = h[p(B) - 0.5]$; $t = k[p(A) - 0.5]$, where h, k are constants between 0 and 1. The state of the population can be plotted as a point with coordinates $p(A), p(B)$. Simulate the population, and plot its changes in state. (This is a diploid genetic version of Dawkins' 'Battle of the sexes' game. The dynamic behaviour is unexpected.)

2. Write a program simulating Levene's model, and use it to check conditions (4.8) for a stable polymorphism. Modify the program to include 'habitat selection' by females: that is, mating is random, but females lay eggs in the patch in which they were born. Does habitat selection make a stable polymorphism more likely?